



## **Affirmed Networks Pooling Licensing Server Installation Guide**

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### **Inside ...**

- Licensing Overview
- Preparing the System for Installation
- Standalone Systems
- Other Administrative Information

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## Preface

This document describes how to install, configure, operate, and maintain the Affirmed Networks Pooling Licensing Server (PLS).

---

**Note:** *The Affirmed Networks Mobile Content Cloud™ is the mobile services software solution that operates on virtual platforms such as blade systems.*

---

## Audience

This guide is written for the Administrator who installs, configures, operates, and maintains the PLS.

## Release Information

See the Release Notes for important information, requirements, limitations, and restrictions that apply to this release.

## How to Use This Guide

This guide contains the following chapters and appendices:

| Chapter/Appendix                                                 | Describes                                           |
|------------------------------------------------------------------|-----------------------------------------------------|
| <a href="#">Chapter 1, Affirmed Networks Licensing</a>           | Affirmed Networks licensing overview                |
| <a href="#">Chapter 2, Preparing the System for Installation</a> | How to prepare the system to install the PLS server |
| <a href="#">Chapter 3, Standalone Systems</a>                    | How to install PLS on a standalone system           |
| <a href="#">Chapter 4, Other Administrative Information</a>      | Additional administrative information               |

## Conventions

This guide uses the following conventions, when applicable:

| Description                                                                                | Convention                                   | Examples                                                                                                   |
|--------------------------------------------------------------------------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Command names, keywords, dialog boxes, buttons, icons, menus                               | <b>Times New Roman bold font</b>             | Use <b>configure_ha</b> to configure...<br>Enter <b>y</b> .<br>To continue, press <b>Enter</b> .           |
| Selecting a menu item                                                                      | <b>Menu =&gt; Option</b>                     | Select <b>Security =&gt; Group Management</b> .                                                            |
| Variable parameters for which you supply values                                            | <i>&lt;courier bold italic blue font&gt;</i> | <b>configure_dr --host &lt;host name&gt;</b>                                                               |
| User input                                                                                 | <b>Courier bold blue font</b>                | <b>cd /opt/Affirmed/NMS/bin</b>                                                                            |
| Screen messages, system output, sample source code                                         | Courier regular blue font                    | Checking for License...                                                                                    |
| Required keywords and arguments in square brackets                                         | <b>[ ]</b>                                   | <b>cluster [id] node [id] reboot</b>                                                                       |
| Optional keywords and arguments in curly brackets                                          | <b>{ }</b>                                   | <b>{cdr-file-name}</b>                                                                                     |
| Keywords separated by the pipe symbol. Requires you to specify only one item from the set. | <b> </b>                                     | Is this correct? <b>[y n]</b>                                                                              |
| Keywords representing a single element in angle brackets                                   | <b>&lt; &gt;</b>                             | <b>&lt;host name&gt;</b>                                                                                   |
| Asterisk after a CLI object indicates a required object                                    | <b>*</b>                                     | <b>name*</b>                                                                                               |
| Variables, book and chapter titles, file path names, new terms, and emphasized text        | <i>Times New Roman italic font</i>           | See the <i>Acuitas User's Guide</i> .<br><i>Workflow</i> is an application running on the network element. |



**CAUTION:** Proceed carefully to avoid possible equipment damage or data loss.

---

**Note:** Additional information that may apply to the subject text.

---

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**Tip:** Helpful suggestions.

---

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## Affirmed Networks Licensing

### Overview

The Affirmed Networks License Management System (LMS) consists of three functional entities:

- Entitlement Generation System (EGS) — Affirmed Networks uses a third-party software application (SafeNet/Gemalto) as the EGS to generate entitlements on a per-customer, deployment, or NE basis.
- Pooling Licensing Server (PLS) — The PLS consists of the SafeNet server application running in a bare metal deployment. The PLS manages service licenses for one or more MCCs.
- Affirmed Networks application software to support license enforcement and licensing features

The licensing feature requires you to install the PLS and configure a disaster recovery PLS. The disaster recovery PLS contains temporary licenses and acts as a standby server in the event that the primary server is not available prior to the grace period expiry.

MCC supports binary licenses which are configured as either On (enabled) or Off (disabled) for each feature or function requiring a license.

MCC uses a Time Enforcement Threshold (TE) to enforce all overage limits and issues an alarm or log when a limit is reached. The TE is set at 150%.

License certificates and revocation are generated from Affirmed Networks order administration and applied to the PLS. For details, see the *Affirmed Networks Pooling License Certificate Generation and Revocation Guide*.

## Licensing CLI Commands

Use the following CLI commands to activate and configure licensing:

```
an3000-mcm-slot7cpu1# license
```

Possible completions:

|                      |                                       |
|----------------------|---------------------------------------|
| add-authorized-mcc   | Add an MCC to the PLS authorized list |
| add-certificate      | Add License Certificate               |
| get-pls-fingerprint  | Get the PLS fingerprint               |
| management-discovery | Force management discovery of PLS     |
| restart-pls          | Restart the PLS                       |
| revoke-certificate   | Revoke License Certificate            |

```
an3000-mcm-slot7cpu1(config)# license server ?
```

Possible completions:

IP Address of PLS Server

Port used by the PLS Server

Once a license server is configured and licenses obtained, the system forces user logouts:

```
**** License state change, ending user session****
```

Upon re-login, users have access to all licensed services.

Use the following CLI commands to display licensing information:

```
an3000-mcm-slot7cpu1# show license
```

Possible completions:

|                 |                                        |
|-----------------|----------------------------------------|
| lm-status       | Display the licensable services status |
| pls-status      | Display the PLS status                 |
| services-status | Display the licensable services status |

| Output modifiers

<cr>

an3000-mcm-slot7cpu1# **show license lm-status <entry>**

|                   |                          |
|-------------------|--------------------------|
| current-pls       | Current PLS              |
| entry             | -                        |
| management-entity | Current PLS              |
| state             | State of License Manager |

an3000-mcm-slot7cpu1# **show license pls-status <ip-address>**

|                   |                           |
|-------------------|---------------------------|
| ip-address        | IP Address                |
| lock-code         | Lock code                 |
| lock-criteria     | Lock criteria             |
| management-entity | Managed entity IP Address |
| management-state  | Managed State             |
| mode              | Mode of PLS               |
| state             | State of PLS              |
| time              | PLS Time                  |
| version           | PLS version               |

an3000-mcm-slot7cpu1# **show license services-status <feature> <version>**

|                |                                    |
|----------------|------------------------------------|
| feature        | licensable feature                 |
| hard-limit     | Hard limit slices for license      |
| license-state  | State of the service               |
| limit          | Soft limit slices for license      |
| mcc-consumed   | Number of slices held by local MCC |
| stat-value     | Current value of stat              |
| total-consumed | Number of slices held by all MCCs  |

|                 |                       |
|-----------------|-----------------------|
| type licensable | service type          |
| unit-size       | Slice size multiplier |
| version         | licensable version    |

## Licensing Statistics

The Affirmed Networks statistics director collects statistics associated with all binary and metric-based data to determine overage and trigger enforcement actions. See *Acuitas Performance Statistics*.

## Licensing Alarms

Service enforcement is indicated through the following alarms:

- **anLicenseCapacityLimit:** Approaching or crossed licensed service capacity limit. Upon reaching x percent of the capacity limit, a major event will be generated. The event will become critical as the limit is crossed.
- **anLicenseCapacityEnforcement:** Approaching or crossed licensed service capacity enforcement level. Upon reaching x percent of the capacity enforcement level, a major event will be generated. The event will become critical as the enforcement level is crossed.
- **anLicenseLifetime:** Approaching or passed licensed service lifetime. Upon reaching x percent of the license lifetime, a major event will be generated. The event will become critical as the lifetime is passed.
- **anLicenseLifetimeEnforcement:** Approaching or passed licensed service enforcement expiration date. Upon reaching x percent of the license expiration date, a major event will be generated. The event will become critical as the expiration is passed.
- **anLicensePlsSeparation:** The MCC has separated from the PLS server and will enter local license management state. During this grace period, the MCC will manage licensed service control based on the last PLS view. Services will be impacted if the server is not restored within the grace period.
- **anLicensePlsGraceExpired:** Connection to the PLS server has not reestablished within the grace period so license-able services are terminated.

## Preparing the System for Installation

To prepare the system to install the PLS server and database software, complete the instructions in the section that applies to your system:

- [Bare Metal System](#)
- [Virtual Machine](#)

### Bare Metal System

This section describes how to prepare a Bare Metal system to install the PLS server and database software. Complete the instructions in the section that applies to your system:

- [HP Proliant DL380 Gen8 System](#)
- [HP Proliant DL380 Gen9 System](#)

## HP Proliant DL380 Gen8 System

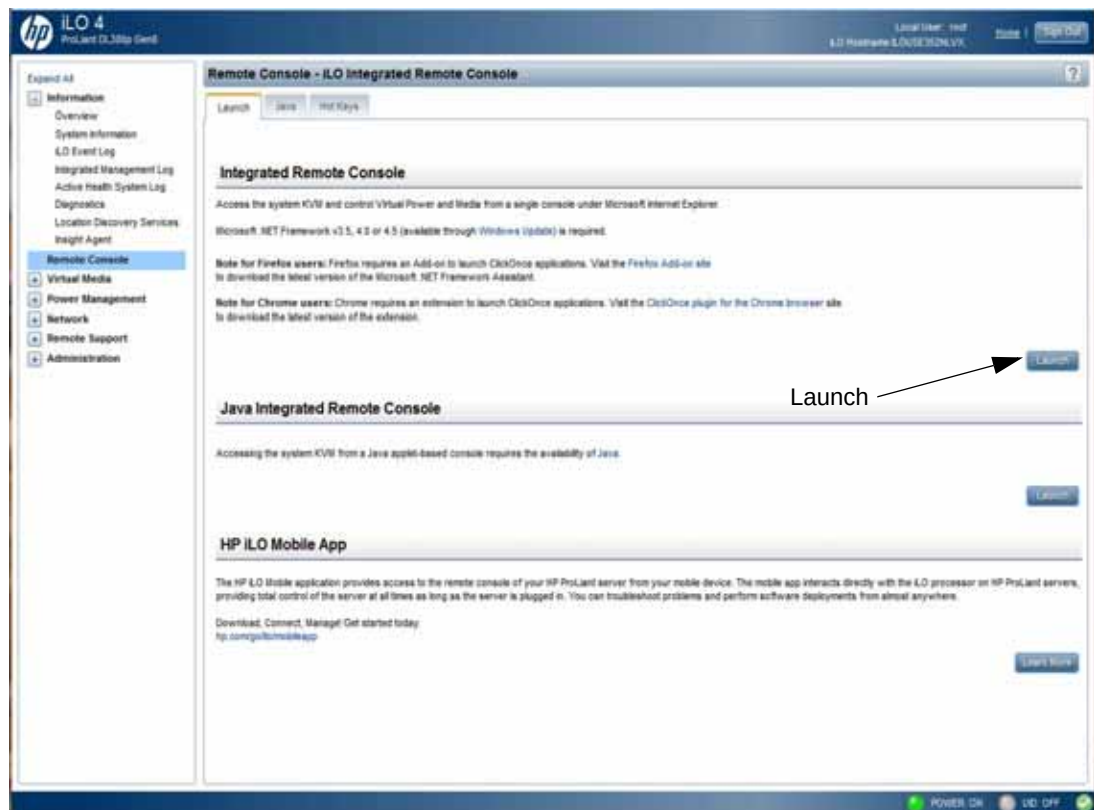
### Initial Procedure to Set Up RAID

For a brand new system, perform this initial procedure to set up RAID. For existing systems, skip to [Procedure to Prepare the Bare Metal System](#).

#### To set up RAID:

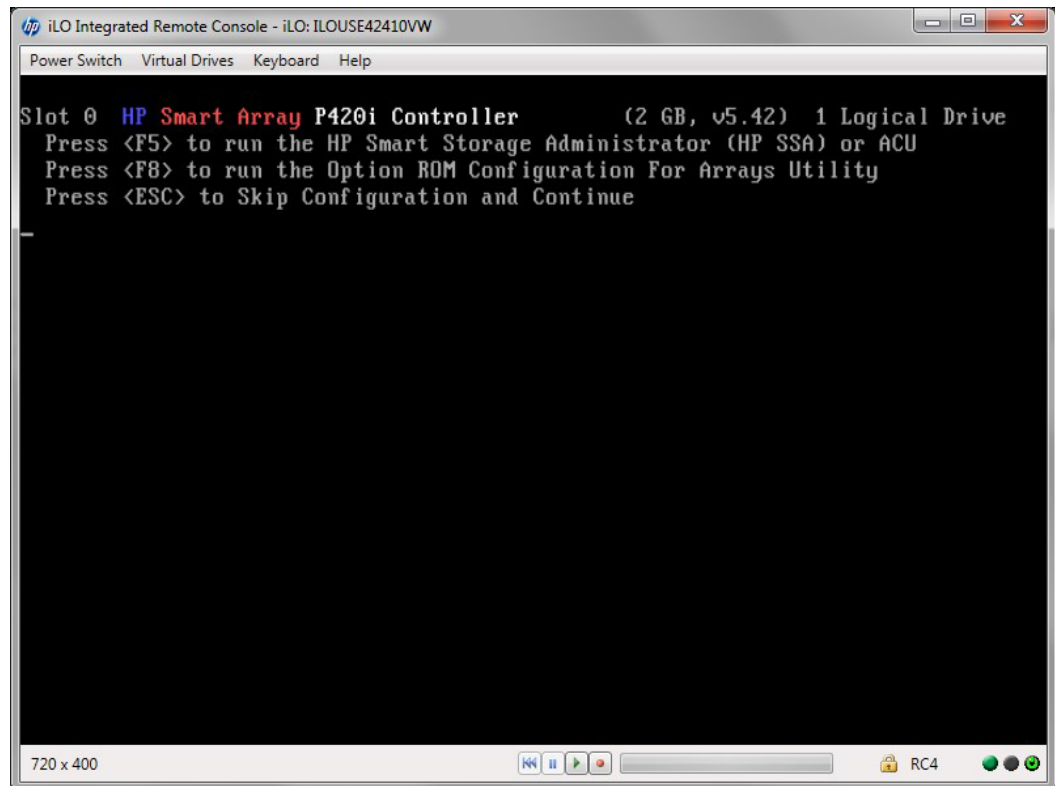
- 1 From a browser, log onto the iLO and click **Launch** to access the Remote Console (Firefox browser requires a Firefox Add-on to launch the Remote Console).

Figure 2-1 iLO



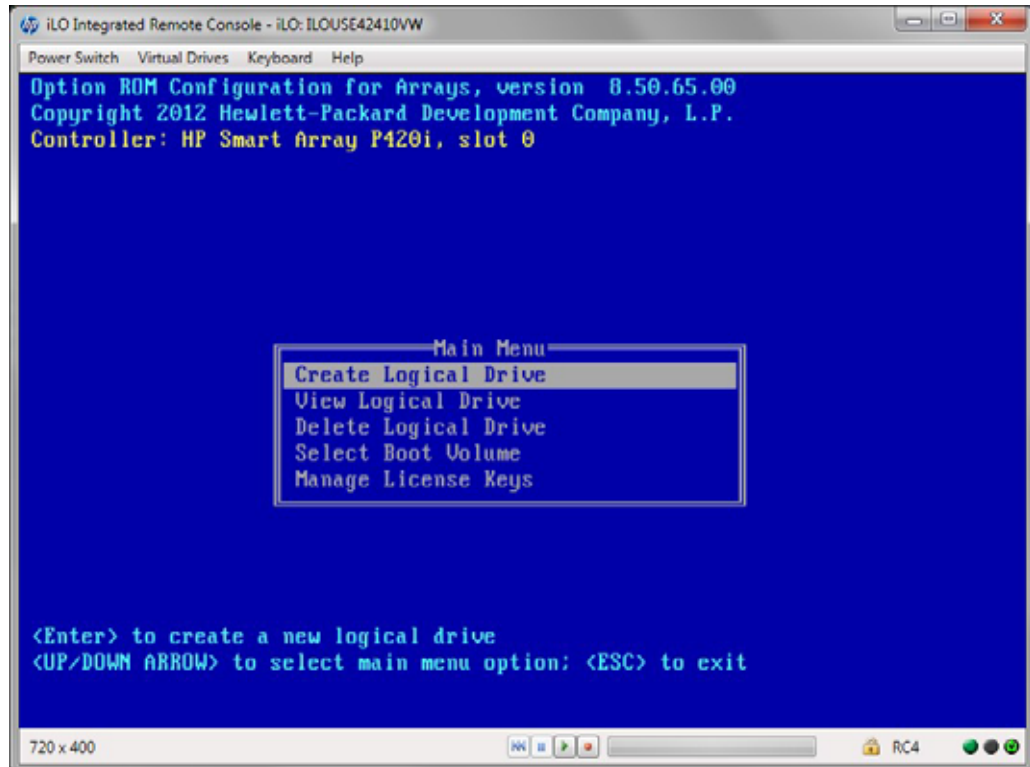
- 2 At the prompt HP Smart Array prompt, press <F8> to run the Option ROM Configuration Utility.

**Figure 2-2 HP Smart Array Prompt**



- 3 From the Main Menu, use the arrow key to highlight **Create Logical Drive** and press **Enter**.

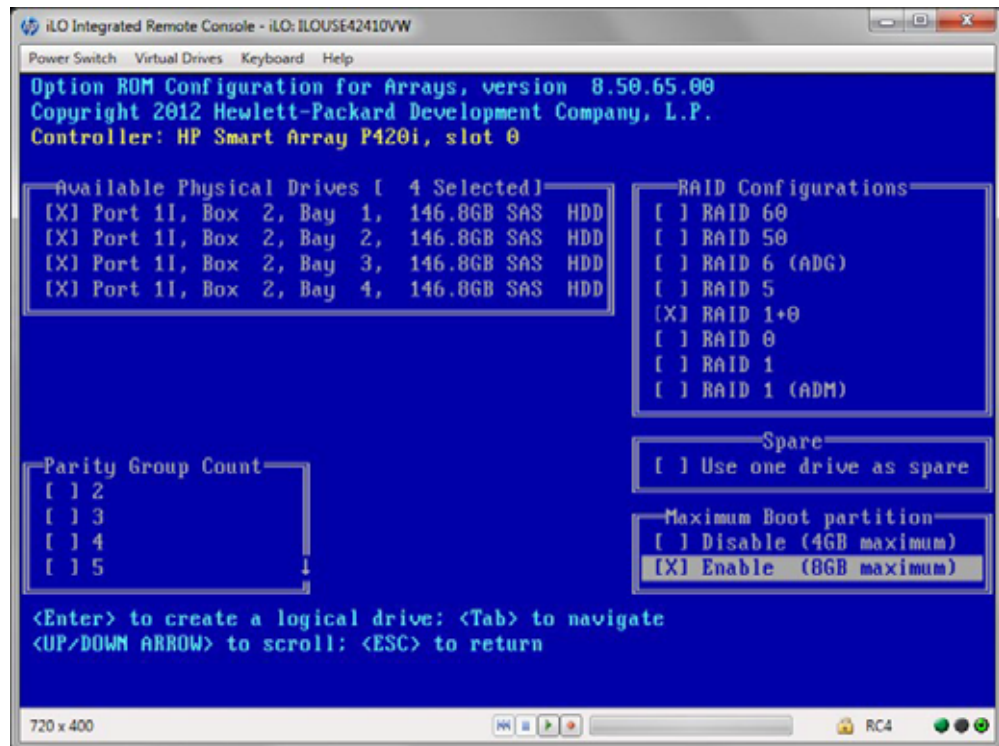
**Figure 2-3 Main Menu**





- 4 Select **RAID 1+0** and **Enable Maximum Boot Partition** and press **Enter** to create the logical drive.

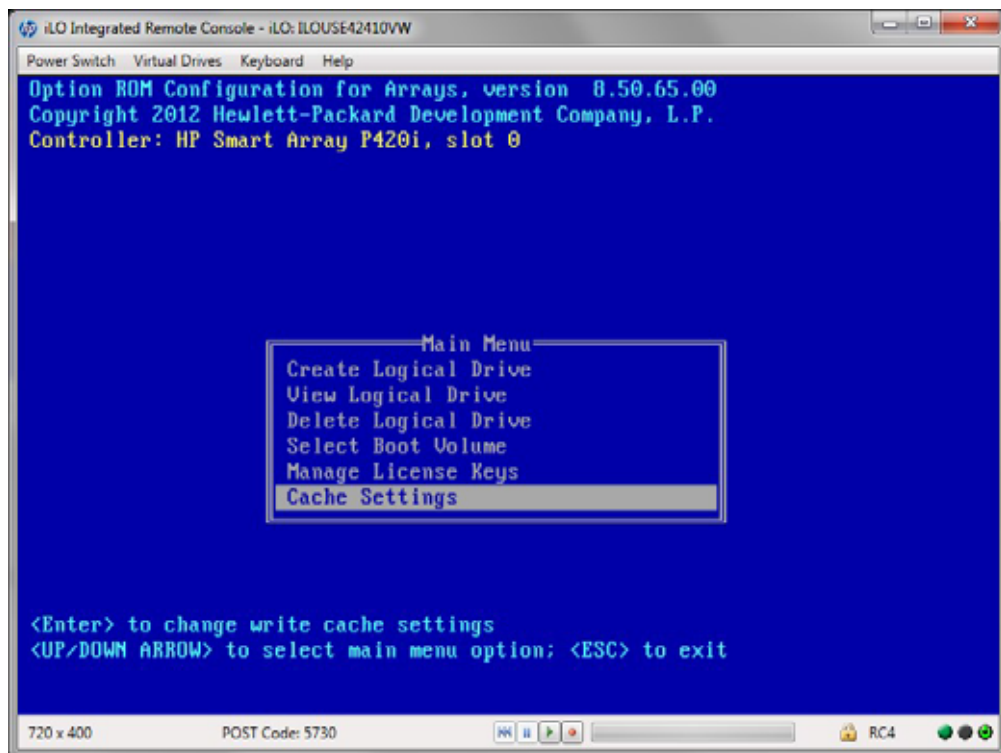
**Figure 2-4 Logical Drive Configuration**



- 5 When prompted, press <F8> to save the configuration.
- 6 When prompted, press **Enter** to continue.

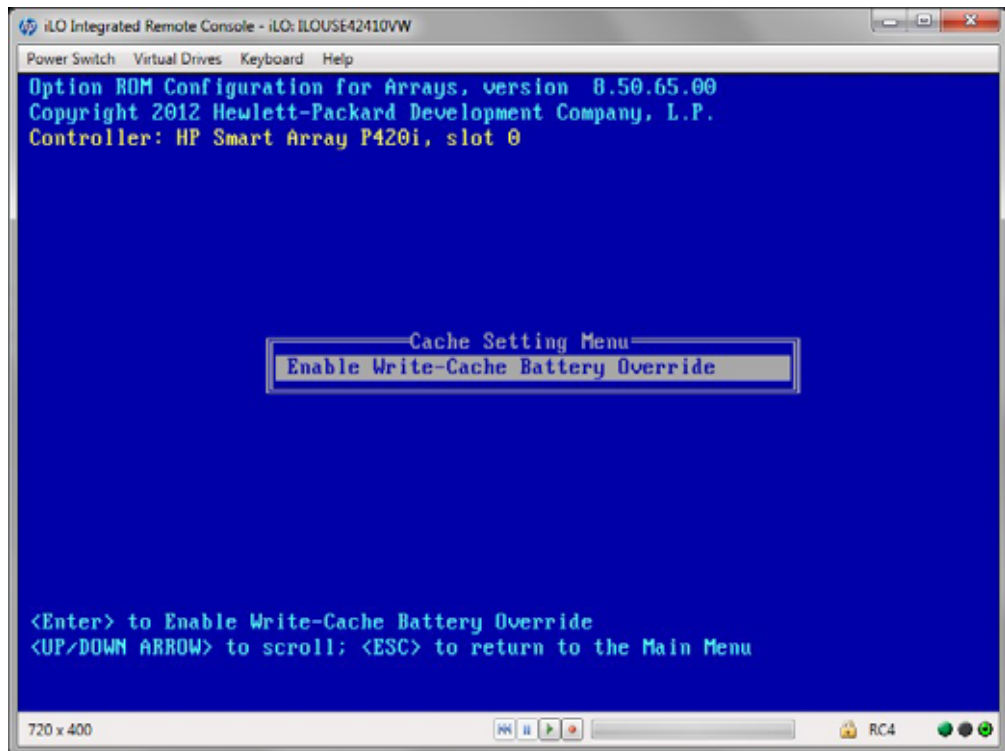
- 7 From the Main Menu, use the arrow key to highlight **Cache Settings** and press **Enter**.

**Figure 2-5 Cache Settings**



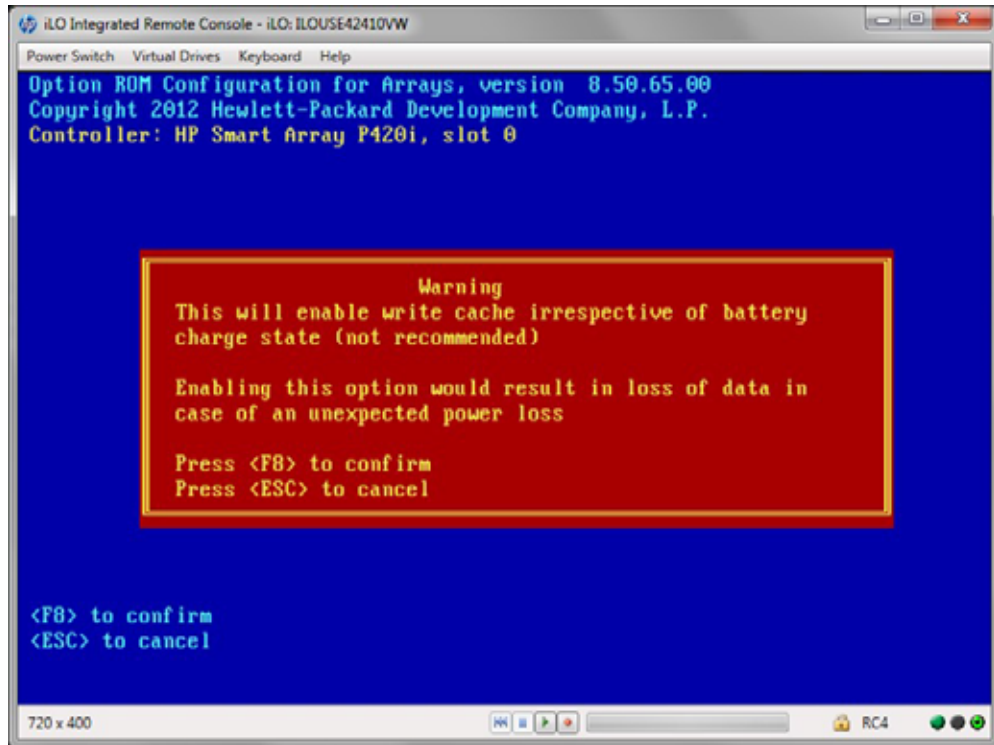
- 8 Use the arrow key to highlight **Enable Write-Cache Battery Override** and press **Enter**.

**Figure 2-6 Enable Write-Cache Battery Override**



- 9 At the warning prompt, press <F8> to confirm.

**Figure 2-7 Warning Prompt**



- 10 When prompted, press **Enter** to continue.

## Procedure to Prepare the Bare Metal System

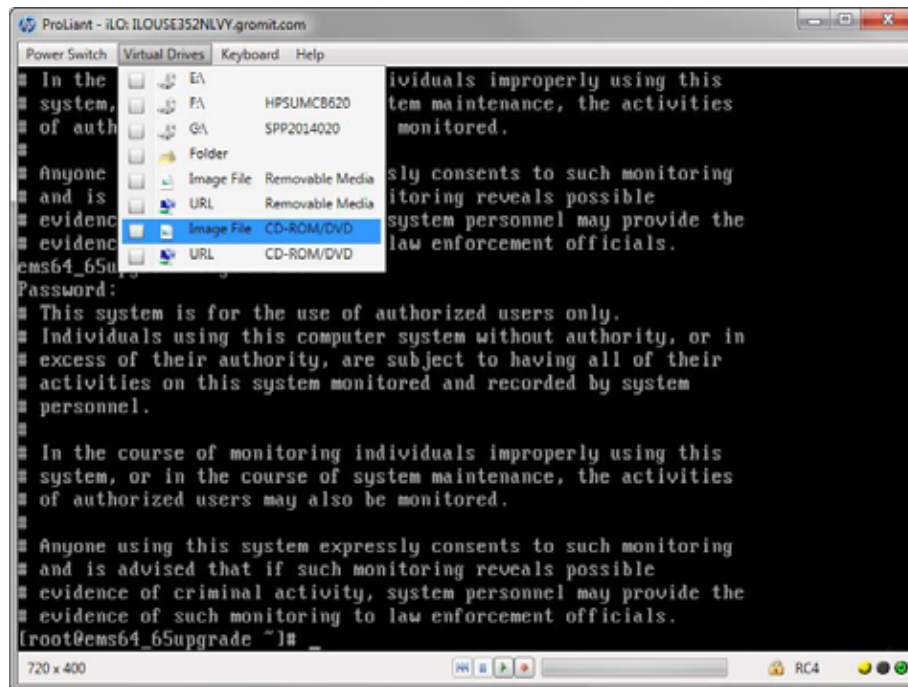
### To prepare a Bare Metal system:

- 1 If the system is already running a version of PLS, backup the database:
  - a Stop the PLS server:
 

```
pls stop
```
  - b Issue the following commands to backup the database:
 

```
pls backup
```
  - c Move the backup to a remote server.
- 2 Download the ISO image from Affirmed Networks and save it to a location accessible from the server to be upgraded (such as, a local computer):
  - HP Proliant DL380 Gen8 system: an-pls-v1.3-rhel-server-6.8-x86\_64.iso
- 3 From the Remote Control console, select **Virtual Drives** > Image File CD/DVD drive from the Main Menu to attach to the ISO image.

Figure 2-8 Remote Control Console

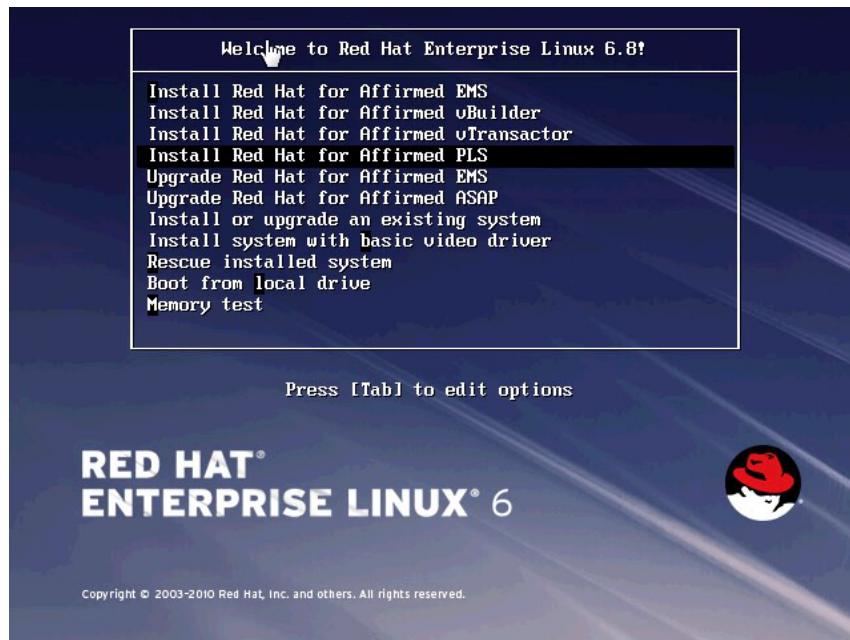


- 4 Enter the following command from the command line:
 

```
reboot
```

It may take several minutes until the HP POST begins.
- 5 From the installer screen, use the arrow key to highlight **Install Red Hat for Affirmed PLS** and press **Tab** to edit options.

Figure 2-9 Installer Screen



- 6 At the **hostname=** prompt, enter the hostname and press **Enter**.

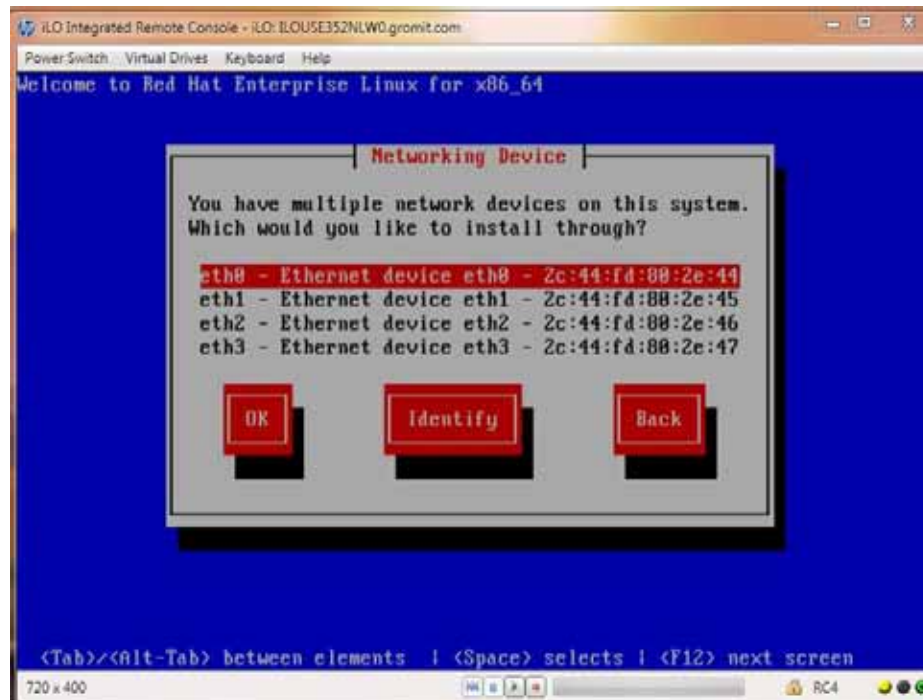
Figure 2-10 hostname= Prompt



It may take several minutes until the Networking Device screen appears.

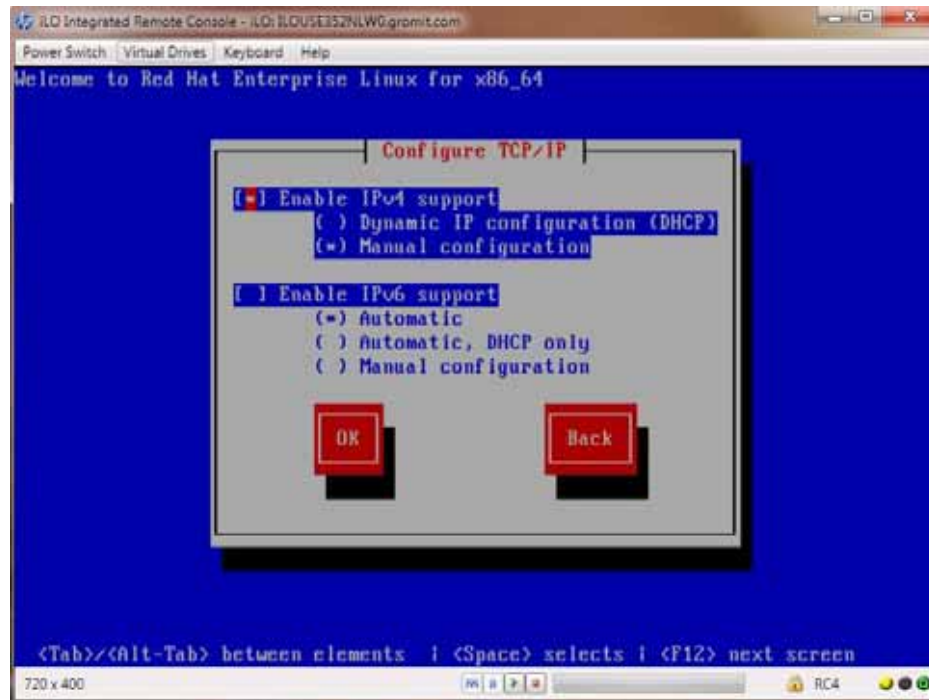
- 7 At the Networking Device screen, use the arrow key to highlight any interface and press the space bar to select it. Then, press **Enter** to continue.

**Figure 2-11 Networking Device**



- 8 At the Configure TCP/IP screen:
  - a Use the arrow key to highlight **Manual Configuration** and press the space bar to select it.
  - b Use the arrow key to highlight **Enable IPv6 support** and press the space bar to deselect it.
  - c Use the arrow key to highlight **OK** and press **Enter**.

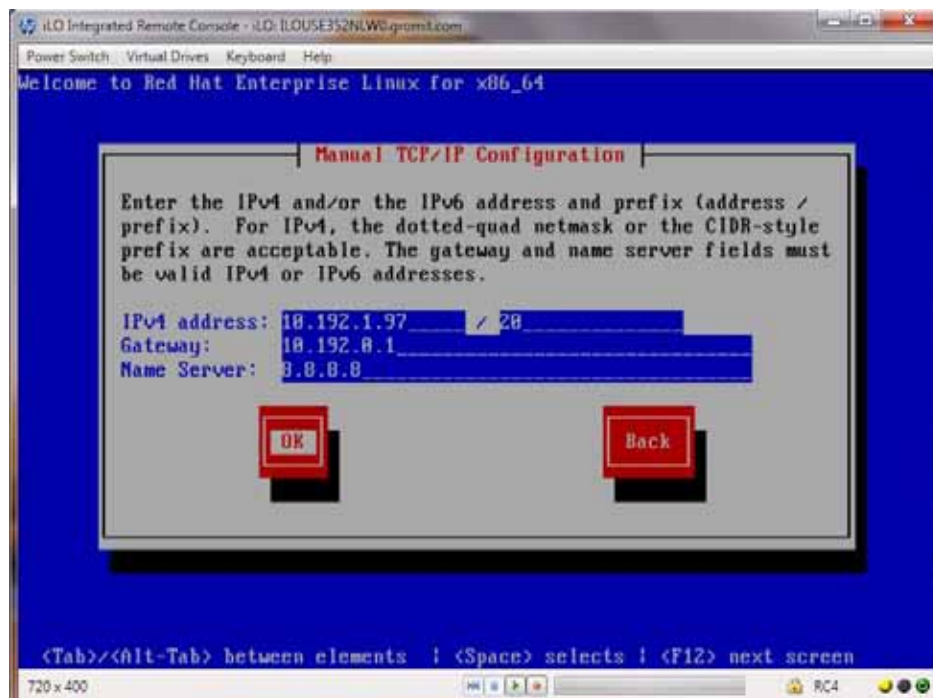
**Figure 2-12 Configure TCP/IP**





- 9 At the Manual TCP/IP Configuration screen, use <Tab> to navigate between the various input lines:
  - a Enter the appropriate IPv4 address and prefix, Gateway, and Name Server. Press the space bar to select each input.
  - b Use <Tab> to select **OK** and press **Enter**.

**Figure 2-13 Manual TCP/IP Configuration**

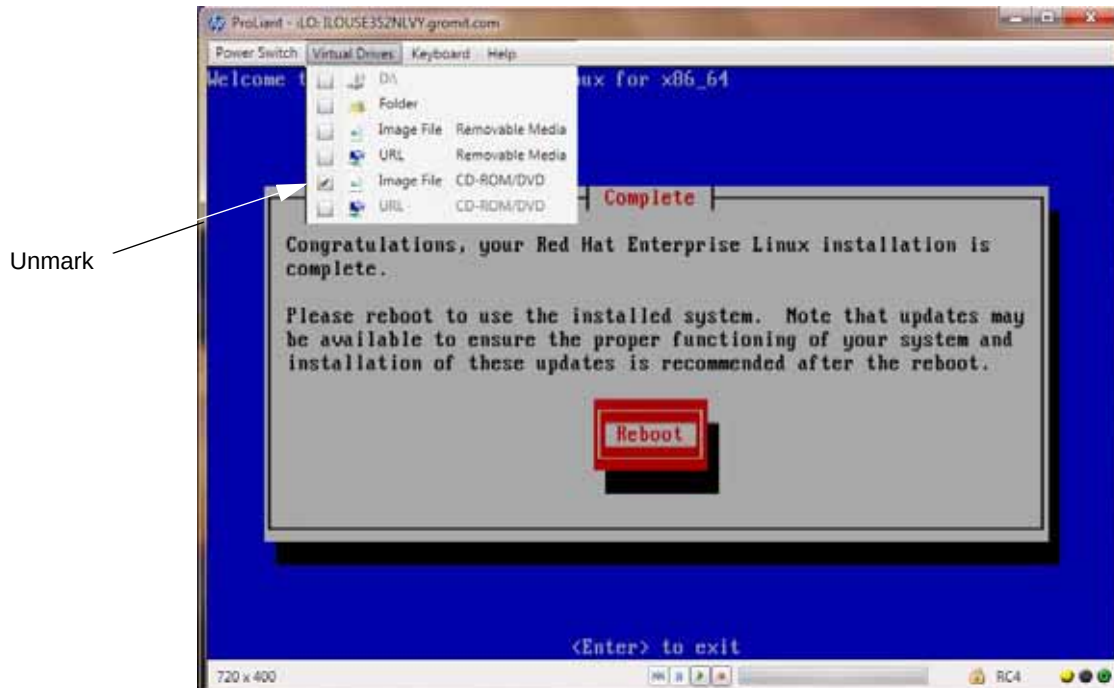


- 10 At the Time Zone Selection screen:
  - a Use <Tab> to highlight **System clock uses UTC** and press the space bar to select it.
  - b Highlight the time zone you want and press the space bar to select it.
  - c Use <Tab> to select **OK** and press **Enter**.

The server installs the RHEL OS (duration: approximately 5-10 minutes).

- 11 When prompted to reboot the system, select **Virtual Drives** from the Main Menu of the Remote Control console and unmark the Image File CD/DVD drive checkbox.

**Figure 2-14 Reboot Prompt**



- 12 Press **Enter** to reboot.

The system is now ready for PLS installation. Proceed to [Standalone Systems](#).

## HP Proliant DL380 Gen9 System

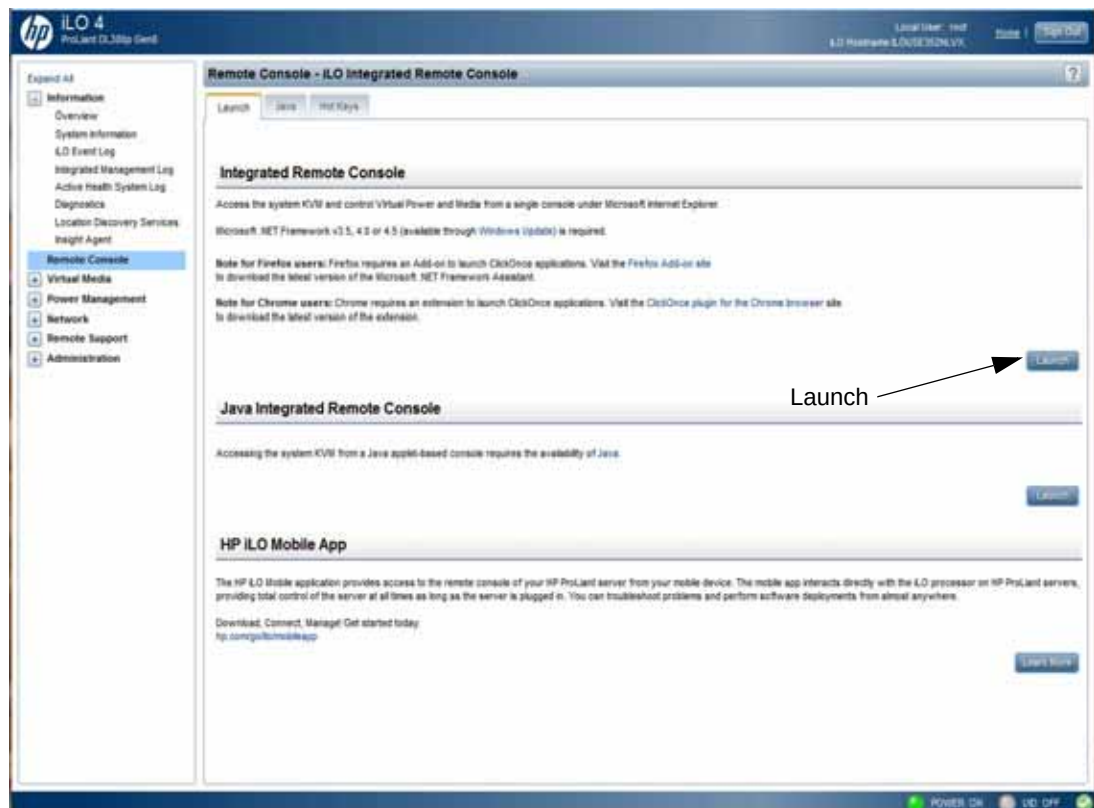
### Initial Procedure to Set Up RAID

For a brand new system, perform this initial procedure to set up RAID. For existing systems, skip to [Procedure to Prepare the Bare Metal System](#).

#### To set up RAID:

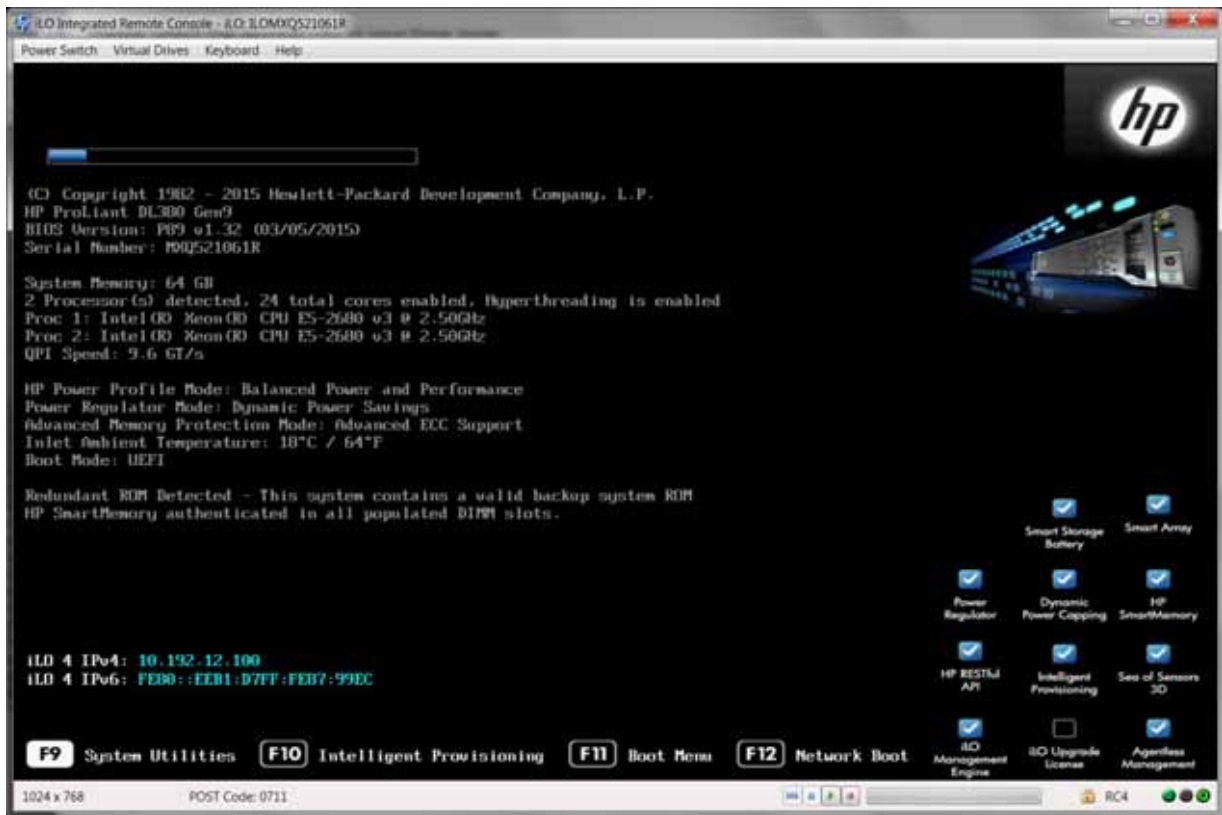
- 1 From a browser, log onto the iLO and click **Launch** to access the Remote Console (Firefox browser requires a Firefox Add-on to launch the Remote Console).

Figure 2-15 iLO



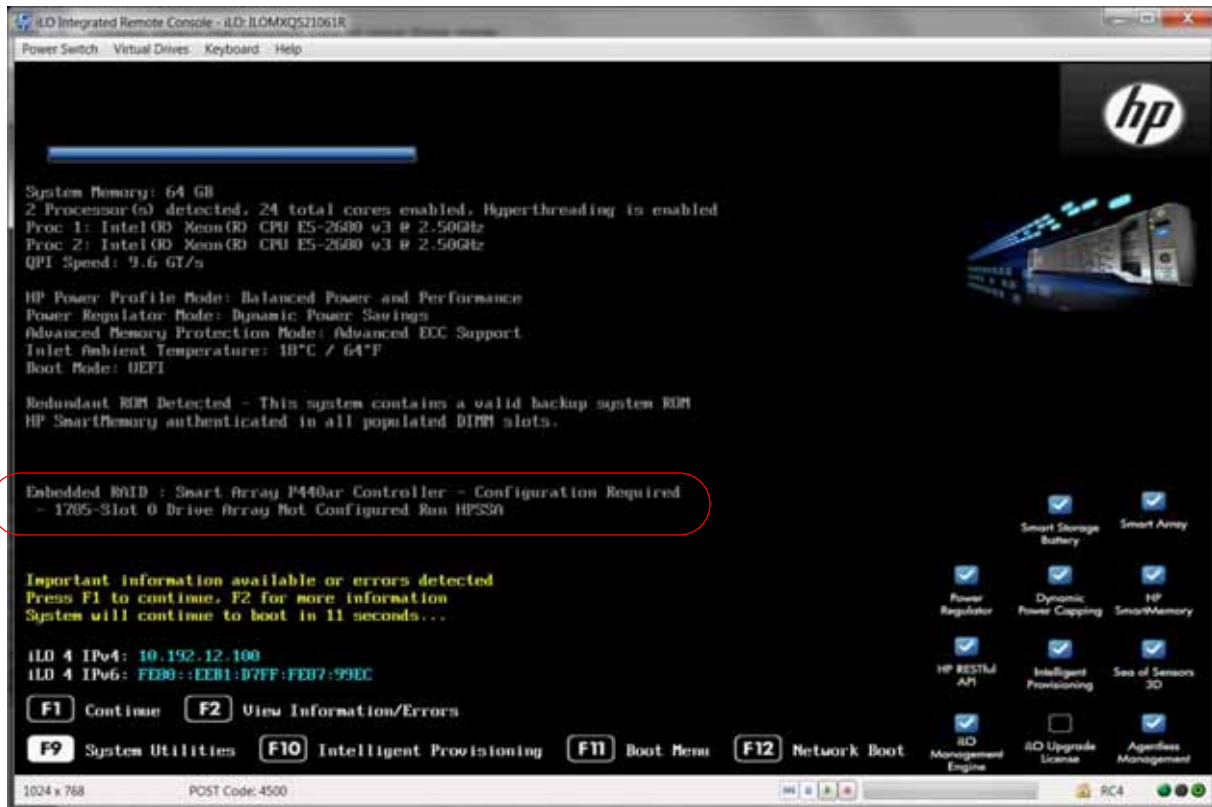
- 2 From the Remote Control console, press <F9> to access System Utilities.

Figure 2-16 Remote Console



- 3 When prompted that RAID has not been set up on the new system, press <F1> to continue.

Figure 2-17 RAID Configuration Required



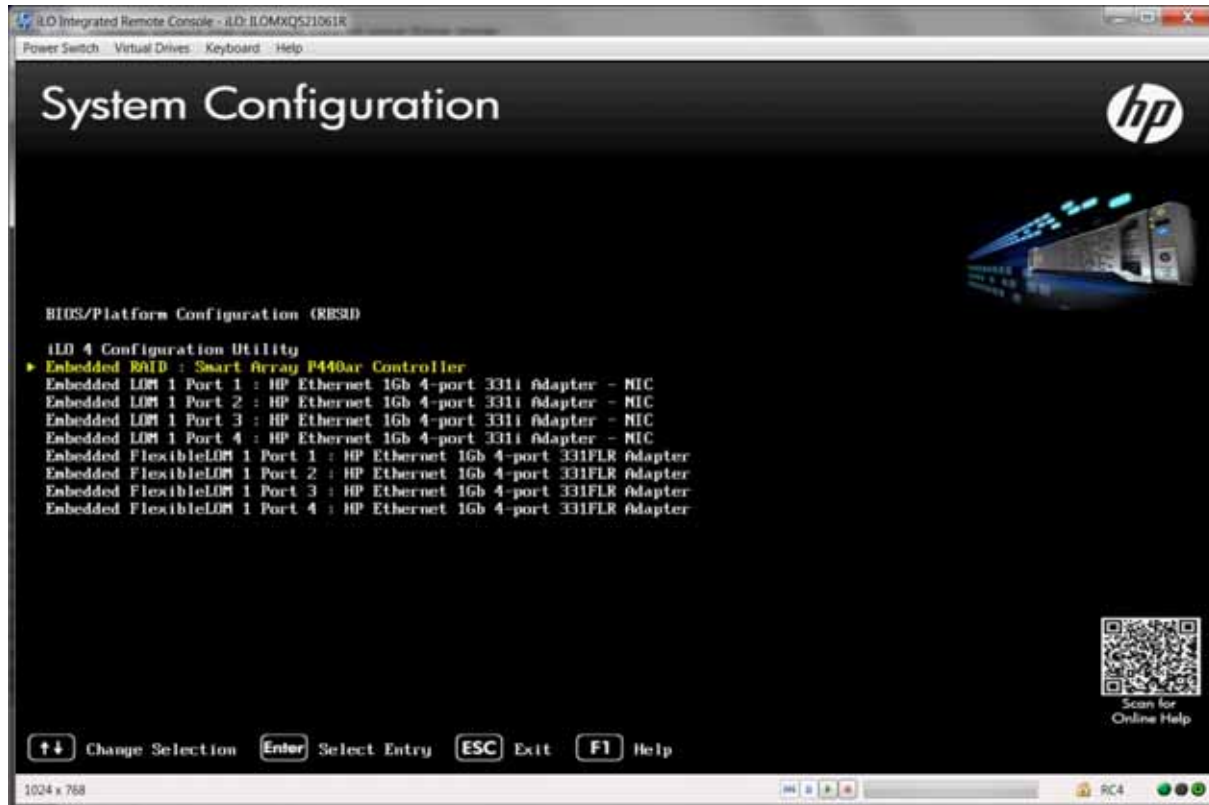
- 4 From the System Utilities menu, use the arrow keys to highlight **System Configuration** and press **Enter**.

**Figure 2-18 System Utilities**



- 5 Use the arrow keys to highlight **Embedded RAID: Smart Array P440ar Controller** and press **Enter**.

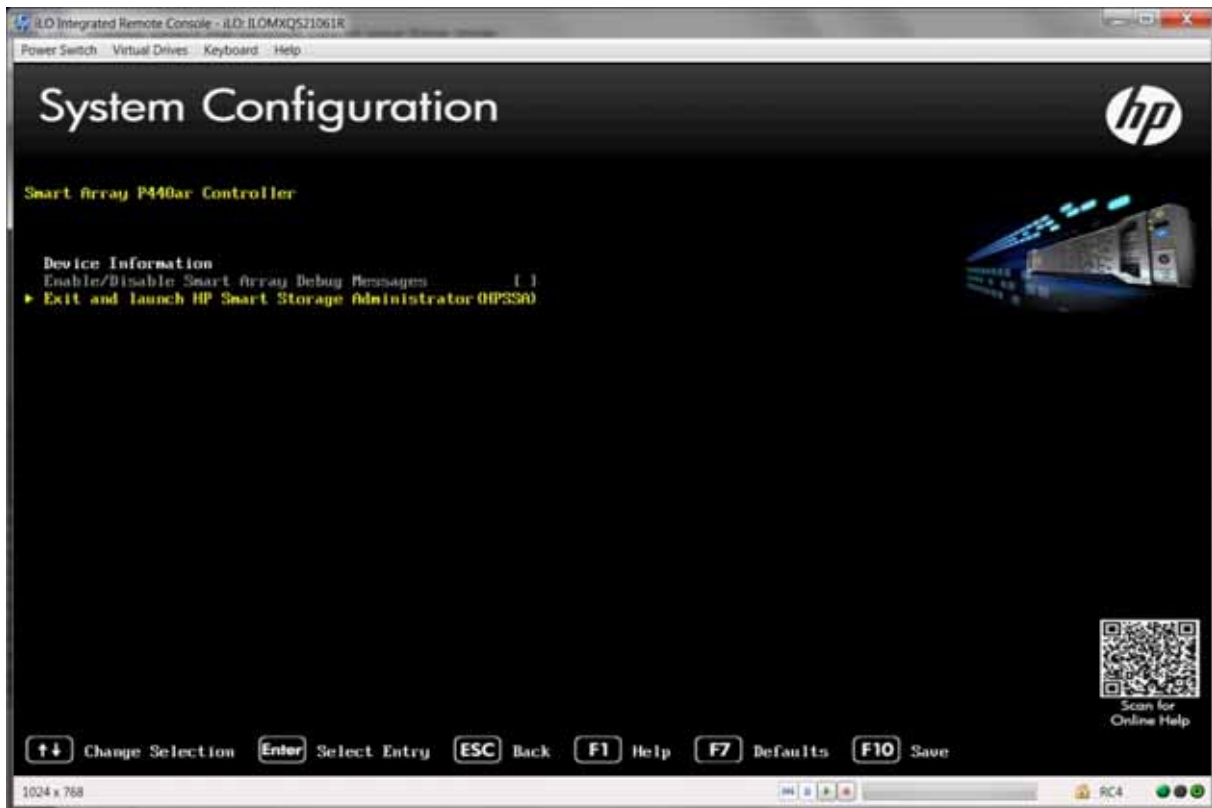
Figure 2-19 System Configuration



- 6 Use the arrow keys to highlight **Exit and launch HP Smart Storage Administrator (HPSSA)** and press **Enter** *only once*.

**Important:** Pressing **Enter** more than once launches HP Intelligent Provisioning, causing you to restart the process.

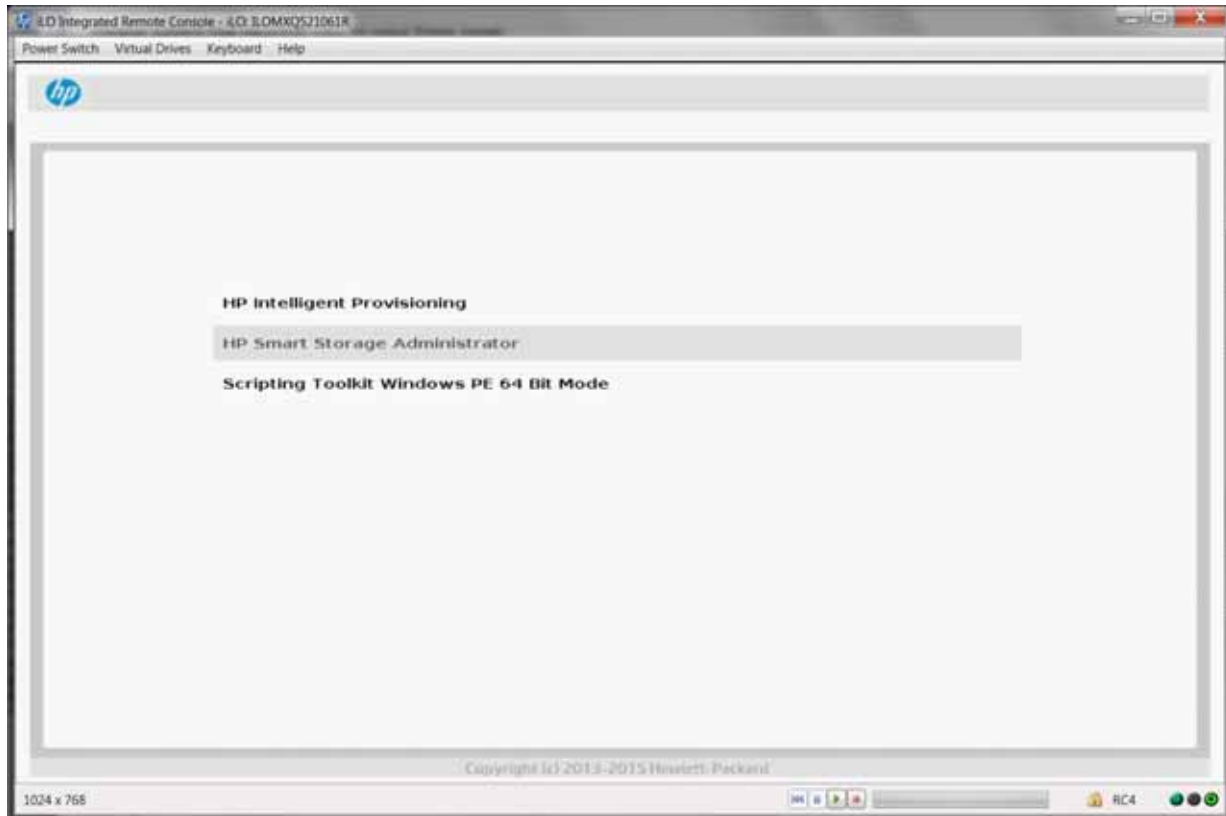
**Figure 2-20 Smart Array P440ar Controller**





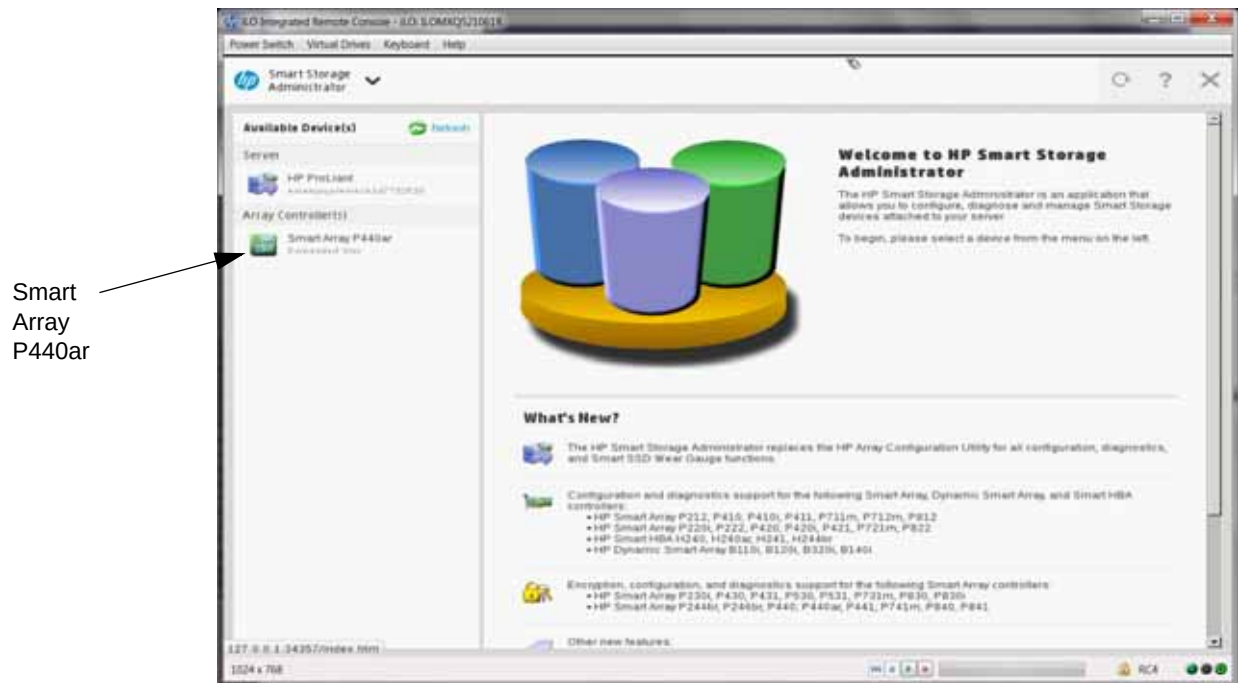
- 7 Use the arrow keys to highlight **HP Smart Storage Administrator** and press **Enter**.  
**Important:** You have only a few seconds to perform this step.

**Figure 2-21 HP Smart Storage Administrator**



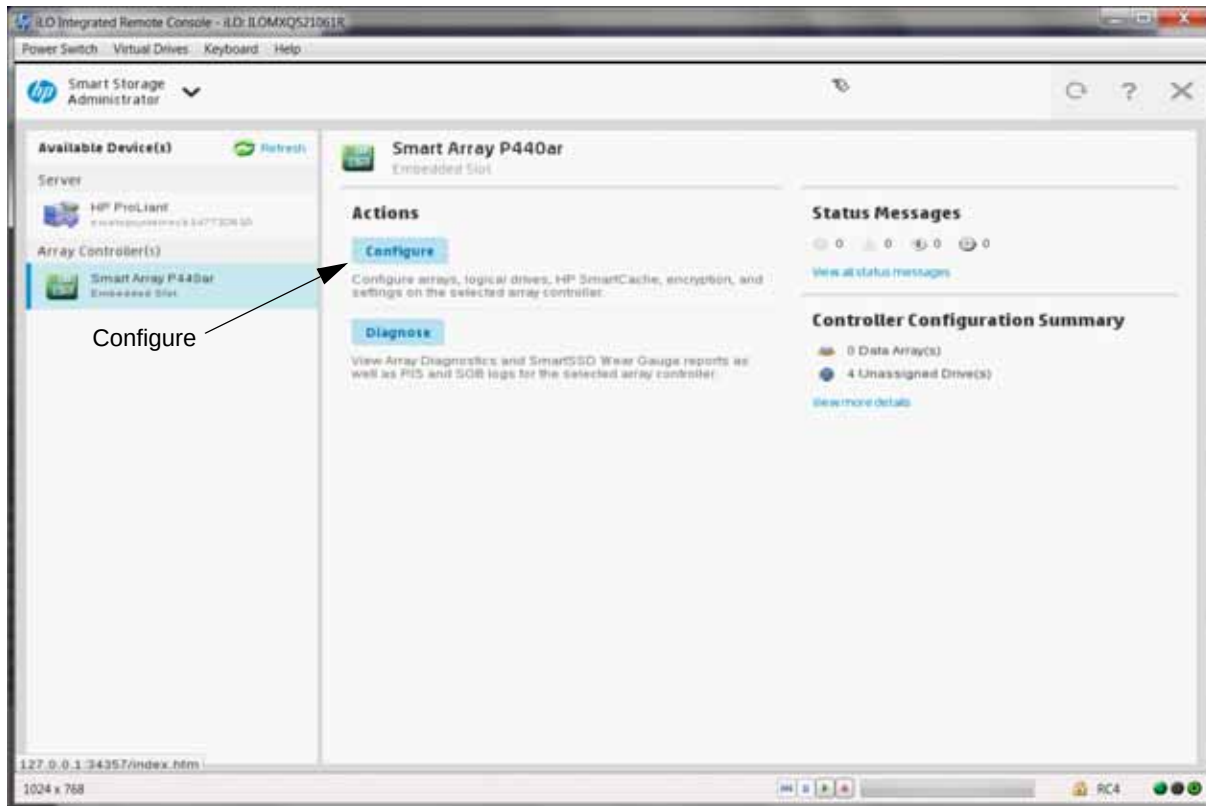
- 8 From the HP Smart Storage Administrator, click **Smart Array P440ar**.

**Figure 2-22 Smart Array P440ar**



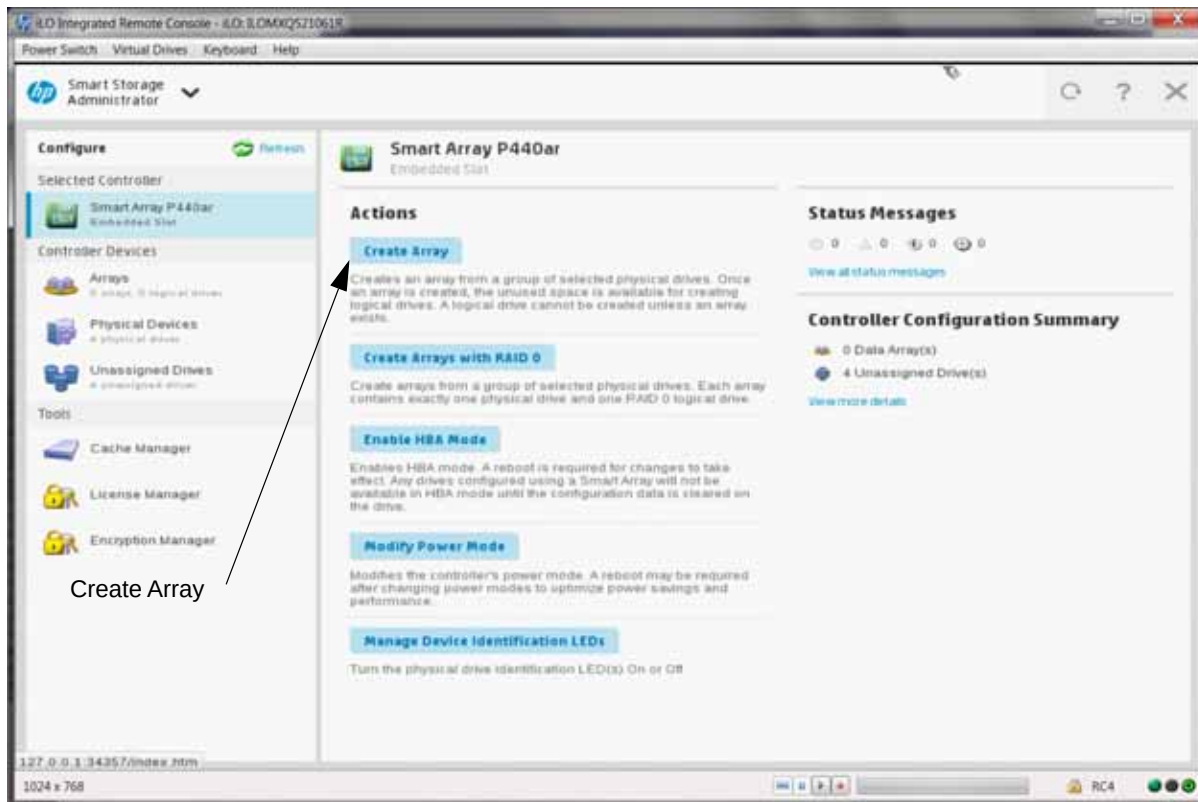
9 Click **Configure**.

Figure 2-23 Configure Smart Array P440ar



10 Click **Create Array**.

Figure 2-24 Create Array



- 11 Mark the **Select All (4)** checkbox and click **Create Array**.

Figure 2-25 Select All (4)

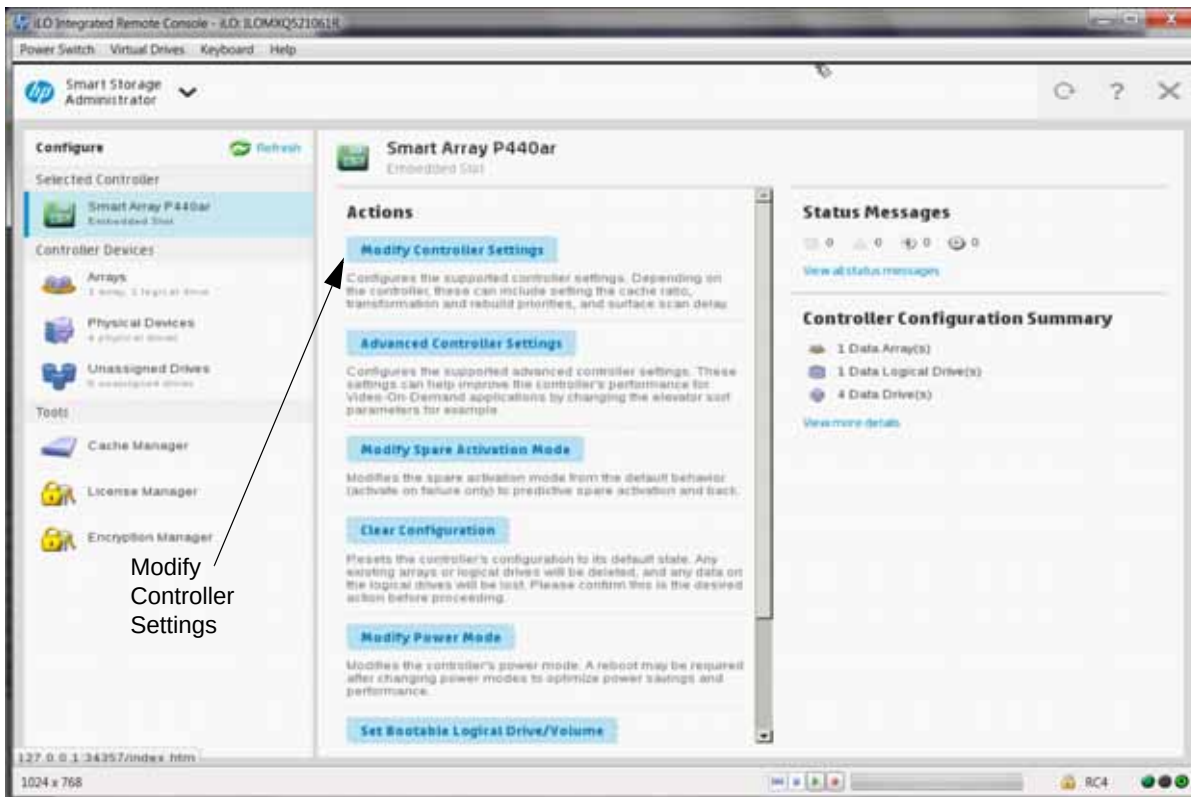


- 12 Mark the following selections and click **Create Logical Drive**.

Figure 2-26 Create Logical Drive Configuration

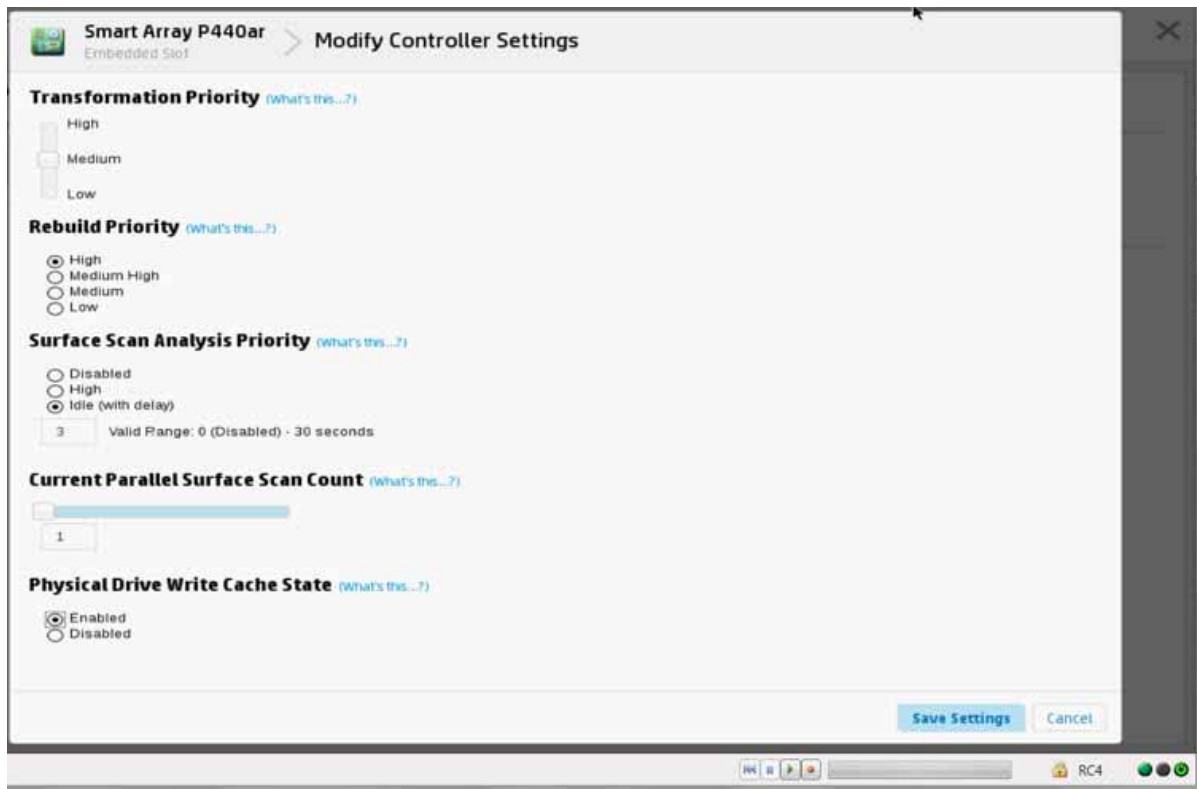


- 13 Click **Finish**.

**14 Click Modify Controller Settings.****Figure 2-27 Modify Controller Settings**

- 15 Mark the following selections and click **Save Settings**.

**Figure 2-28 Controller Settings**

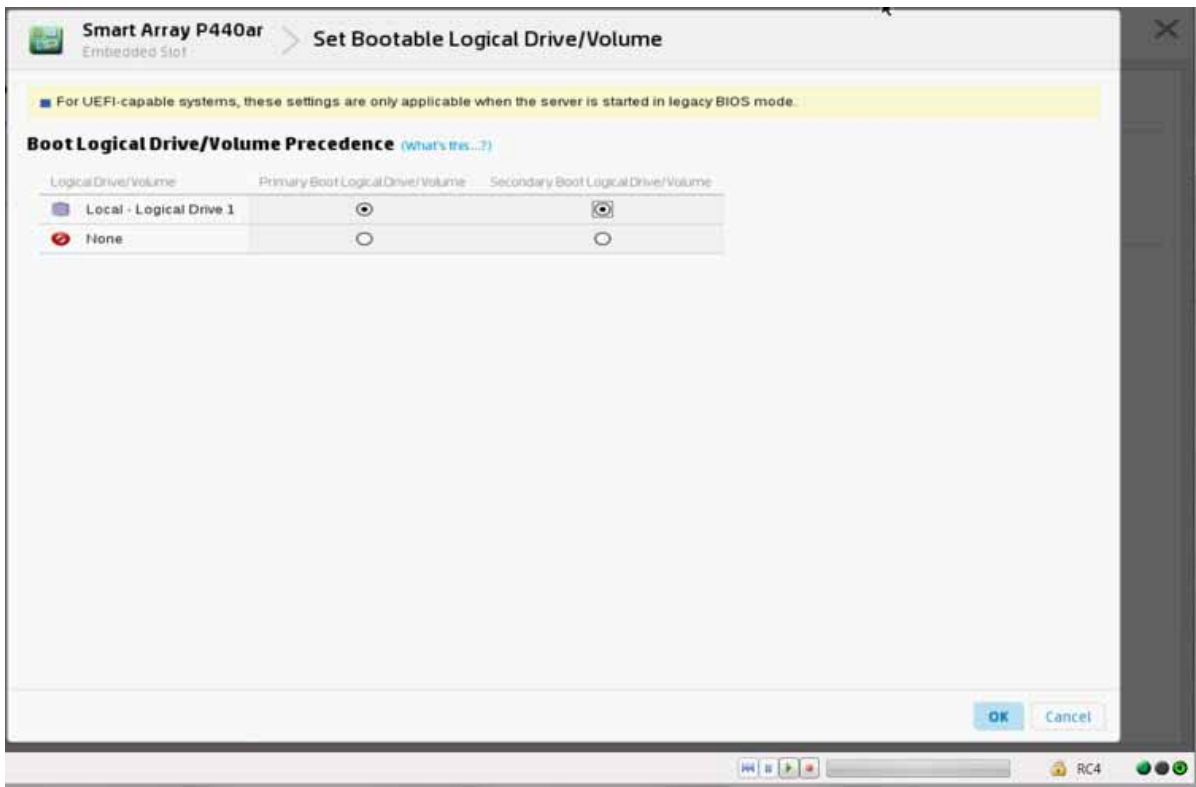


- 16 Click **Finish**.
- 17 Click **Set Bootable Logical Drive/Volume**.



- 18 Mark the following selections and click **OK**.

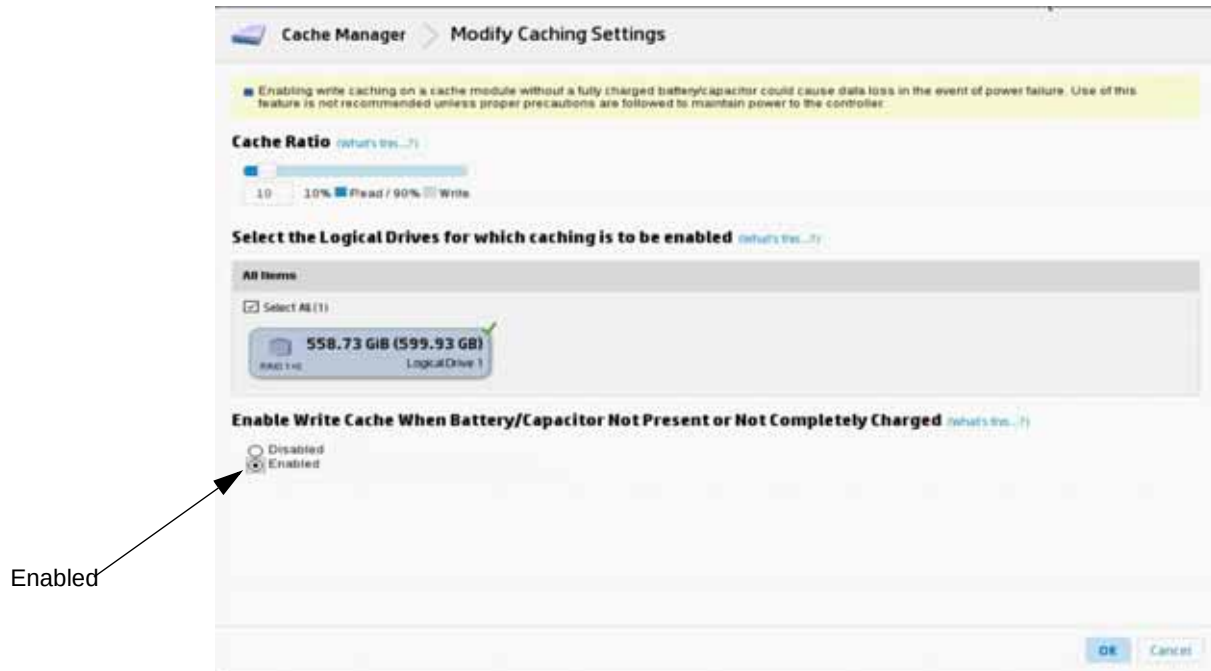
**Figure 2-29 Bootable Logical Drive/Volume Setting**



- 19 Click **Finish**.
- 20 Under **Tools**, click **Cache Manager**.

- 21 Under **Enable Write Cache When Battery/Capacitor Not Present or Not Completely Charged**, mark the Enabled checkbox and click **OK**.

Figure 2-30 Modify Caching Settings



- 22 Click **Finish**.

### ***Procedure to Set Boot Order***

**To set boot order:**

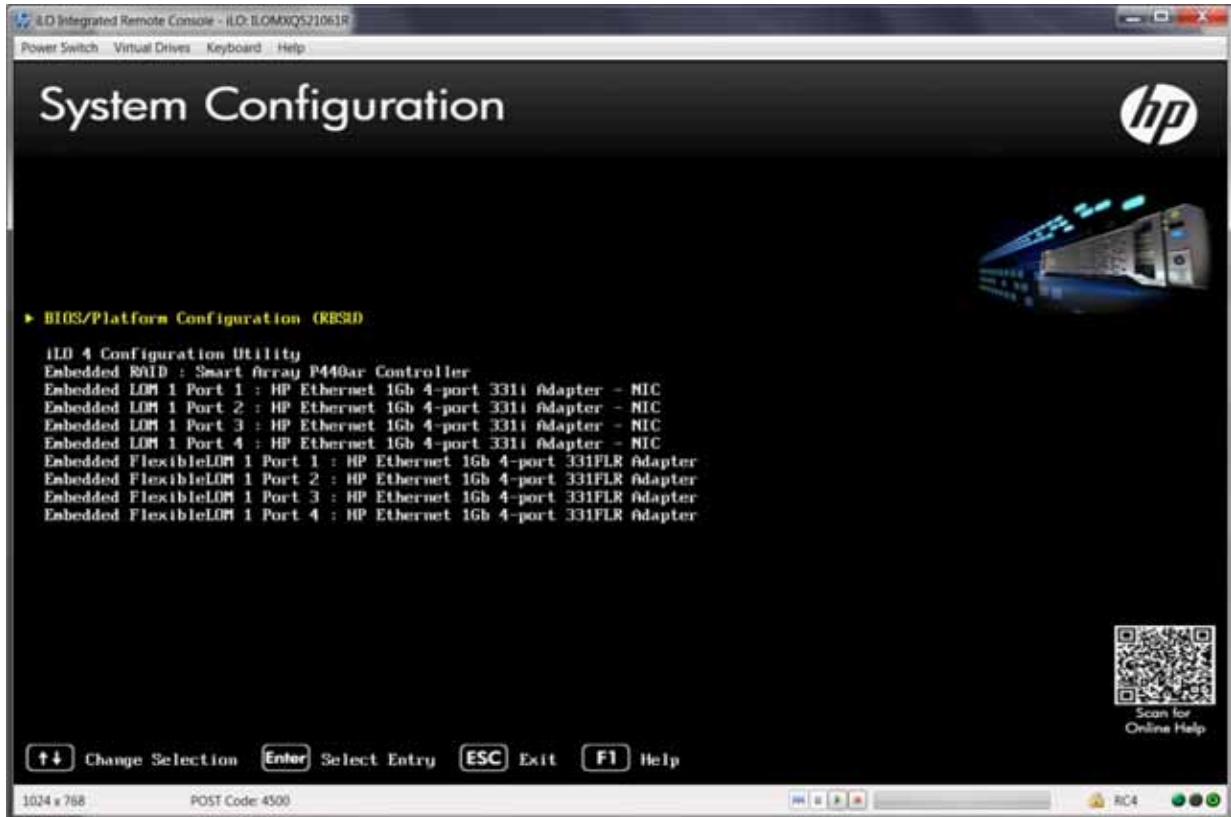
- 1 From the Remote Control console, press <F9> to access System Utilities.
- 2 From the System Utilities menu, use the arrow keys to highlight **System Configuration** and press **Enter**.

**Figure 2-31 System Utilities**



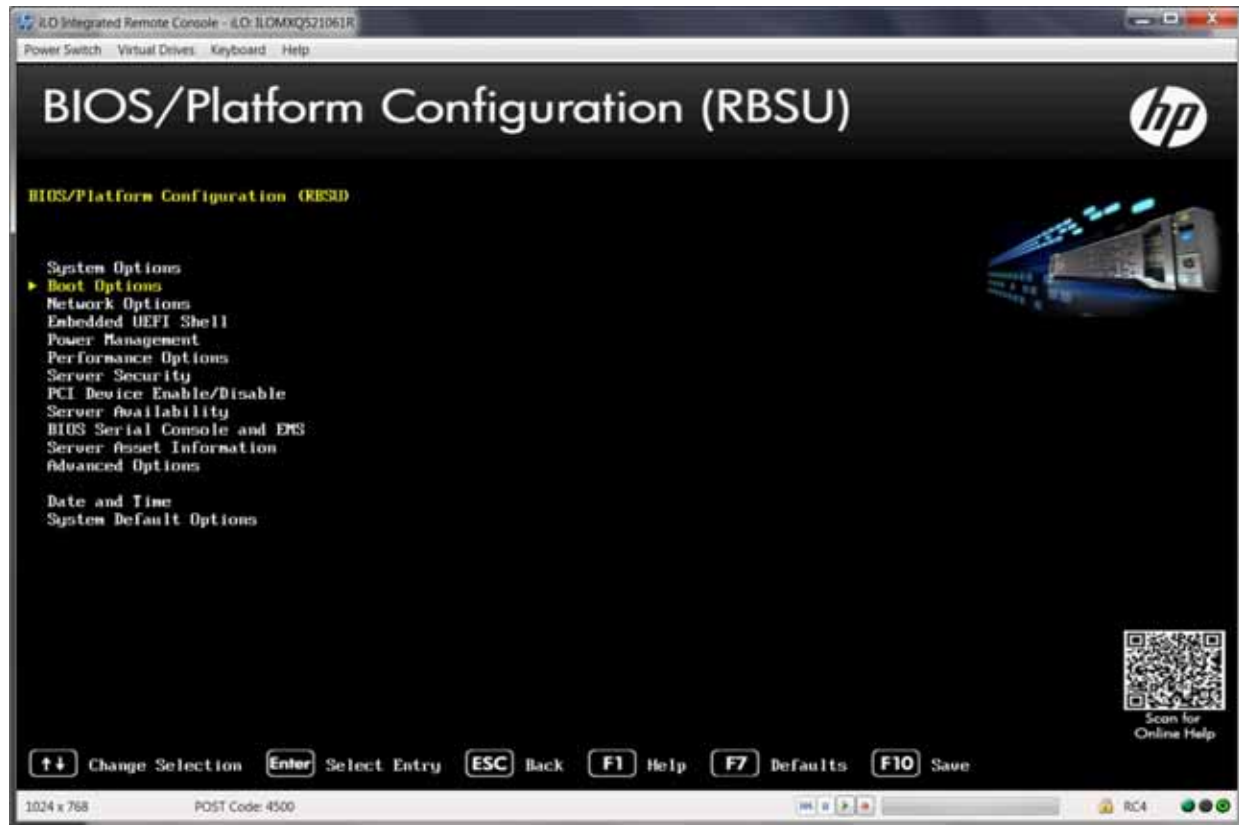
- 3 Use the arrow keys to highlight **Select BIOS/Platform Configuration (RBSU)** and press **Enter**.

Figure 2-32 System Configuration



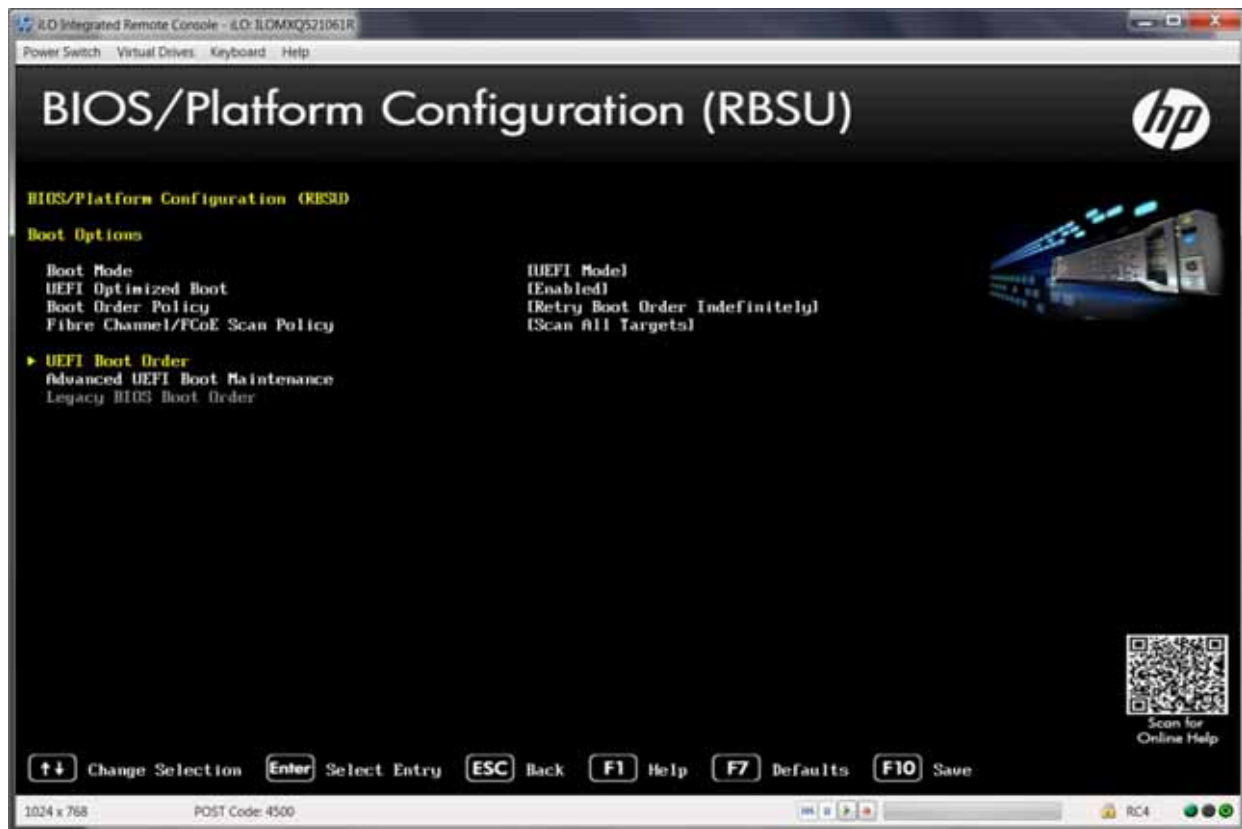
- 4 Use the arrow keys to highlight **Boot Options** and press **Enter**.

Figure 2-33 BIOS/Platform Configuration (RBSU)



- 5 Use the arrow keys to highlight **UEFI Boot Order** and press **Enter**.

Figure 2-34 UEFI Boot Order



- 6 Use the arrow keys to highlight **Embedded RAID**. Use <+> to move the Embedded RAID to the top of the order and press **Enter**.
- 7 At the confirmation prompt to save pending changes, enter **Y**.

## Procedure to Prepare the Bare Metal System

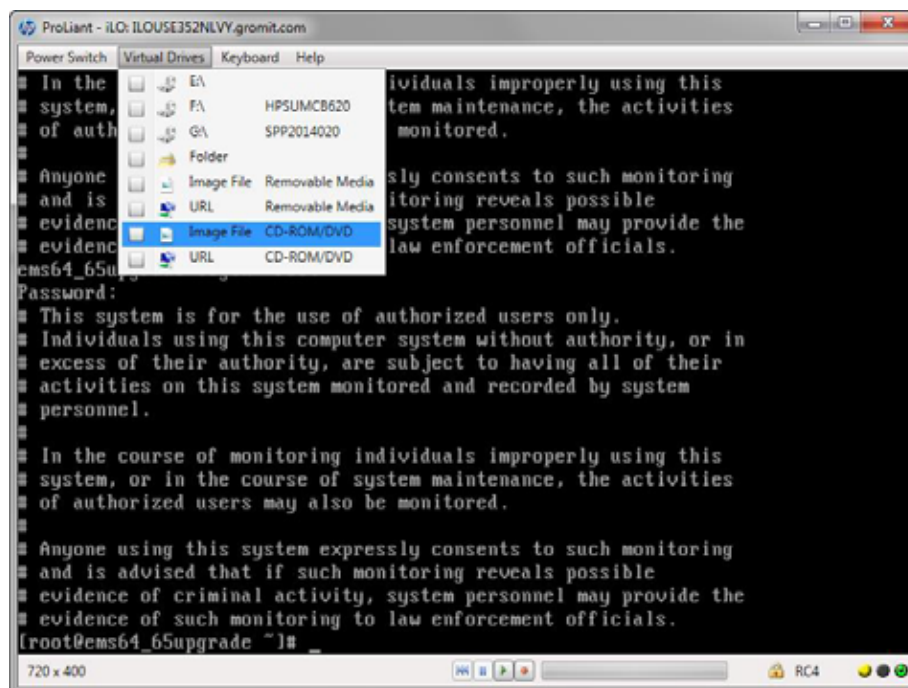
### To prepare a Bare Metal system:

- 1 If the system is already running a version of PLS, backup the database:
  - a Stop the PLS server:
 

```
pls stop
```
  - b Issue the following commands to backup the database:
 

```
pls backup
```
  - c Move the backup to a remote server:
- 2 Download the ISO image from Affirmed Networks and save it to a location accessible from the server to be upgraded (such as, a local computer):
  - HP Proliant DL380 Gen9 system: an-pls-v1.3-rhel-server-6.8-x86\_64.iso
- 3 From the Remote Control console, select **Virtual Drives** > Image File CD/DVD drive from the Main Menu to attach to the ISO image.

Figure 2-35 Remote Control Console



- 4 From the Remote Control console, press <F9> to access System Utilities.
- 5 Use the arrow keys to highlight **One-Time Boot Menu** and press **Enter**.

Figure 2-36 System Utilities



- 6 From the Boot Menu, use the arrow keys to highlight **iLO Virtual USB 2** and press **Enter**.

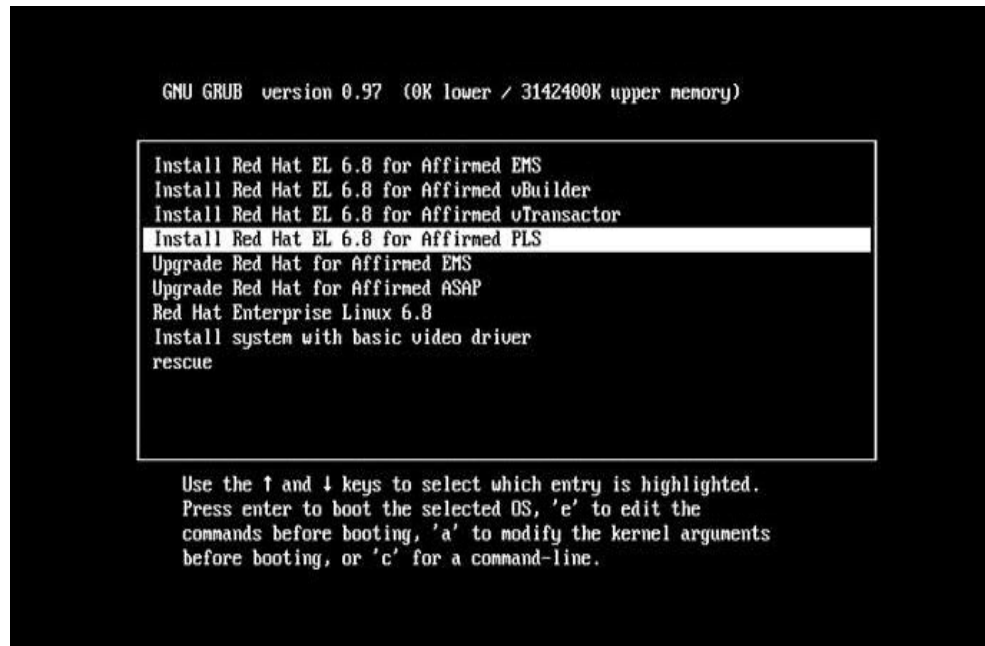
Figure 2-37 Boot Menu





- 7 From the GRUB menu, use the arrow key to highlight **Install Red Hat EL 6.8 for Affirmed PLS**.

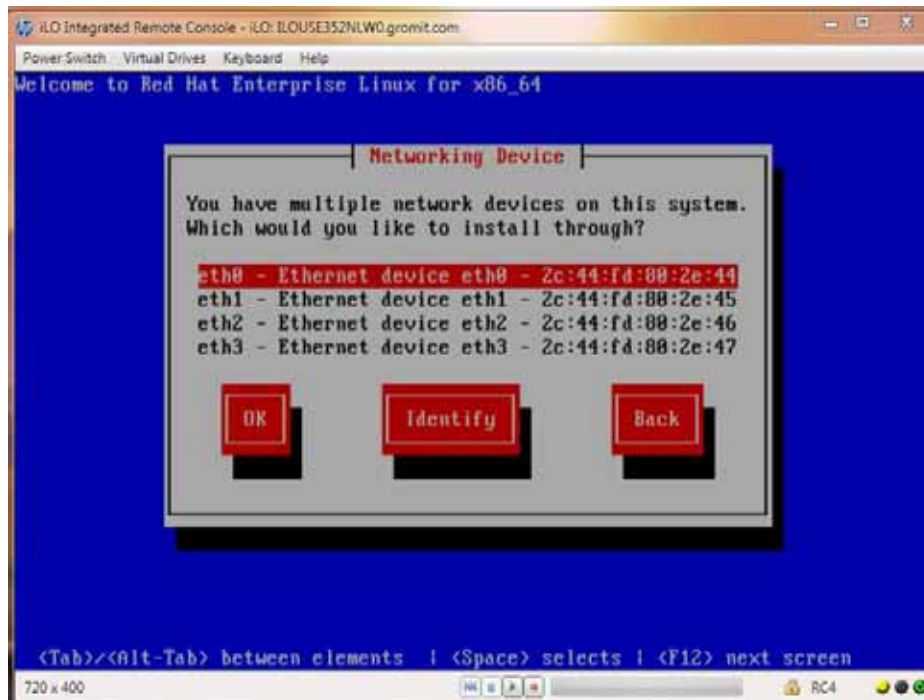
**Figure 2-38 GRUB Menu**



- 8 From the GRUB menu, to edit the hostname, type **<e>** twice.
- 9 Enter the hostname and press **Enter**.
- 10 Type **<b>** to continue boot up.

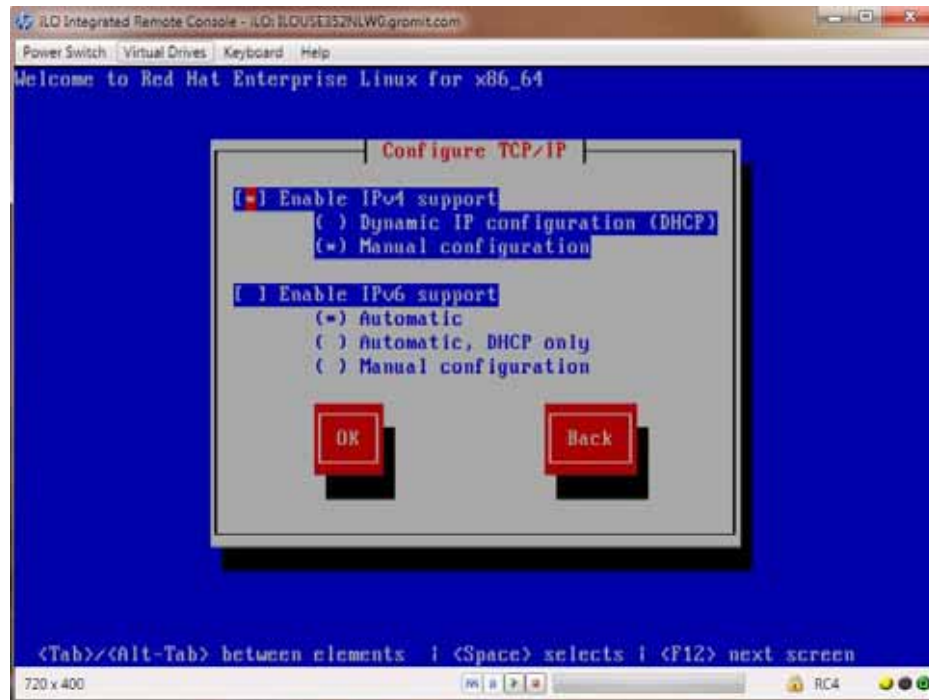
- 11 At the Networking Device screen, use the arrow key to highlight any interface and press the space bar to select it. Then, press **Enter** to continue.

**Figure 2-39 Networking Device**



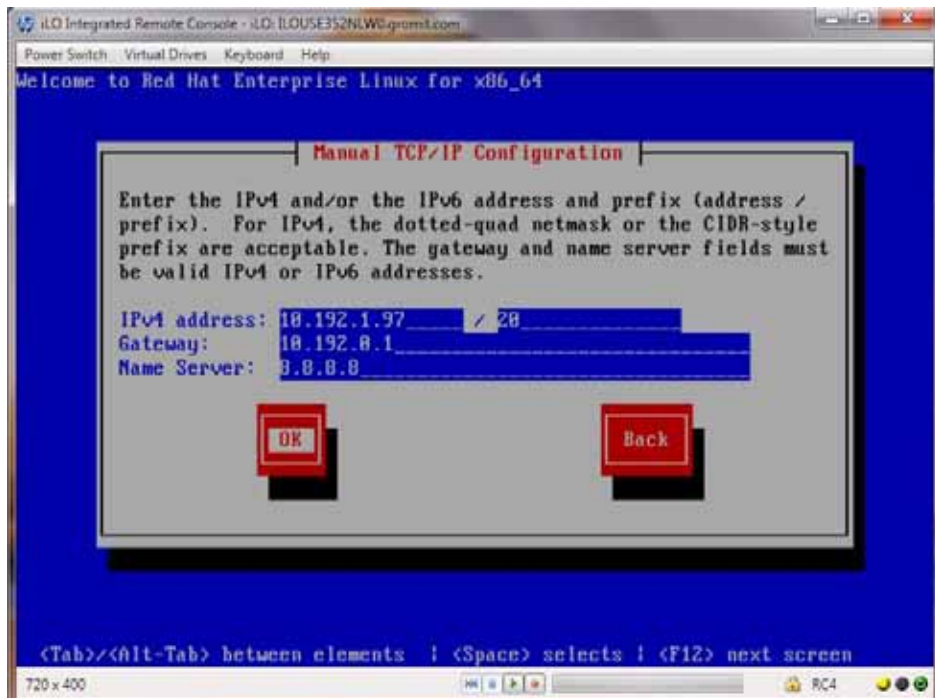
- 12 At the Configure TCP/IP screen:
- Use the arrow key to highlight **Manual Configuration** and press the space bar to select it.
  - Use the arrow key to highlight **Enable IPv6 support** and press the space bar to deselect it.
  - Use the arrow key to highlight **OK** and press **Enter**.

**Figure 2-40 Configure TCP/IP**



- 13 At the Manual TCP/IP Configuration screen, use <Tab> to navigate between the various input lines:
  - a Enter the appropriate IPv4 address and prefix, Gateway, and Name Server. Press the space bar to select each input.
  - b Use <Tab> to select **OK** and press **Enter**.

Figure 2-41 Manual TCP/IP Configuration

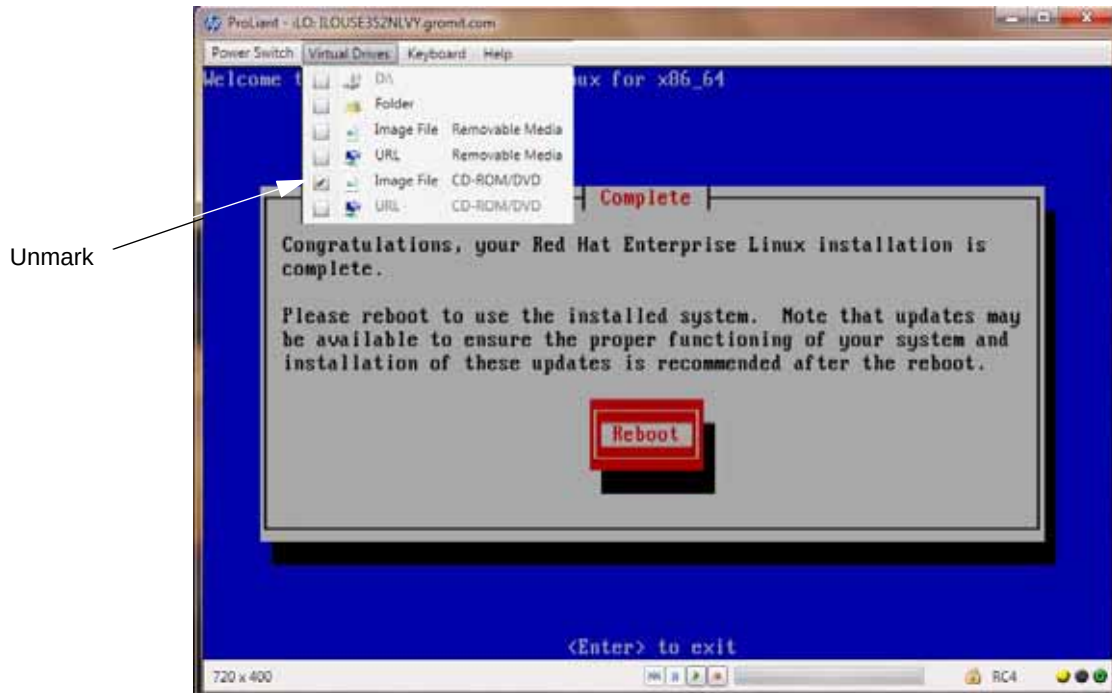


- 14 At the Time Zone Selection screen:
  - a Use <Tab> to highlight **System clock uses UTC** and press the space bar to select it.
  - b Highlight the time zone you want and press the space bar to select it.
  - c Use <Tab> to select **OK** and press **Enter**.

The server installs the RHEL OS (duration: approximately 5-10 minutes).

- 15 When prompted to reboot the system, select **Virtual Drives** from the Main Menu of the Remote Control console and unmark the Image File CD/DVD drive checkbox.

**Figure 2-42 Reboot Prompt**



- 16 Press **Enter** to reboot.

The system is now ready for PLS installation. Proceed to [Standalone Systems](#).

## Virtual Machine

The PLS server can run on either of the following virtual machines:

- Virtual Machine on the HP ProLiant DL380 Server
- Virtual Machine on a blade of the HP BladeSystem c7000

The following table lists the hardware specifications running on a virtual machine.

**Table 2-1 Hardware Specifications**

| Specification   | Description                                                                                                                                                                                                                                    |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hardware</b> |                                                                                                                                                                                                                                                |
| Disk Size       | 1.3 GB<br><i>Note: If your hard disk has more storage available, you can increase the provisioned size once the PLS package has been deployed.</i>                                                                                             |
| CPU             | 2 vCPU<br><i>Note: If your system has the resources available, you can increase the number of CPUs once the PLS package has been deployed.</i><br>For instructions, see <a href="#">Increasing PLS Server Resources on a Virtual Machine</a> . |
| Memory          | 2 GB                                                                                                                                                                                                                                           |

Complete the instructions in the section that applies to your system:

- [ESXi System](#)
- [KVM System](#)

## ESXi System

This section describes how to prepare an ESXi virtual machine, including:

- [Install the OVA Template Package](#)
- [Configure Security Profile](#)
- [Configure the PLS VM to Automatically Start Up](#)
- [Complete the Interface and DNS Setup](#)

This procedure requires you to have VMware ESXi installed and configured, such as enabling NTP on the host. For instructions, see the *MCC Deployment Guide* for your virtual platform.

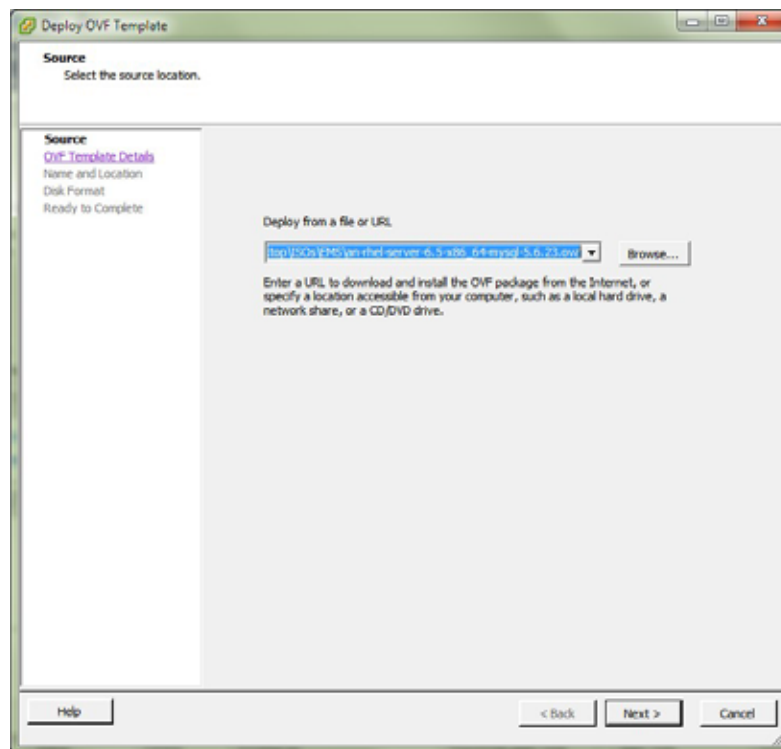
## Install the OVA Template Package

To run the PLS server on a virtual machine, you must install the Red Hat and MySQL Open Virtualization Archive (OVA) template package on the server before you install PLS.

### To install the OVA template package:

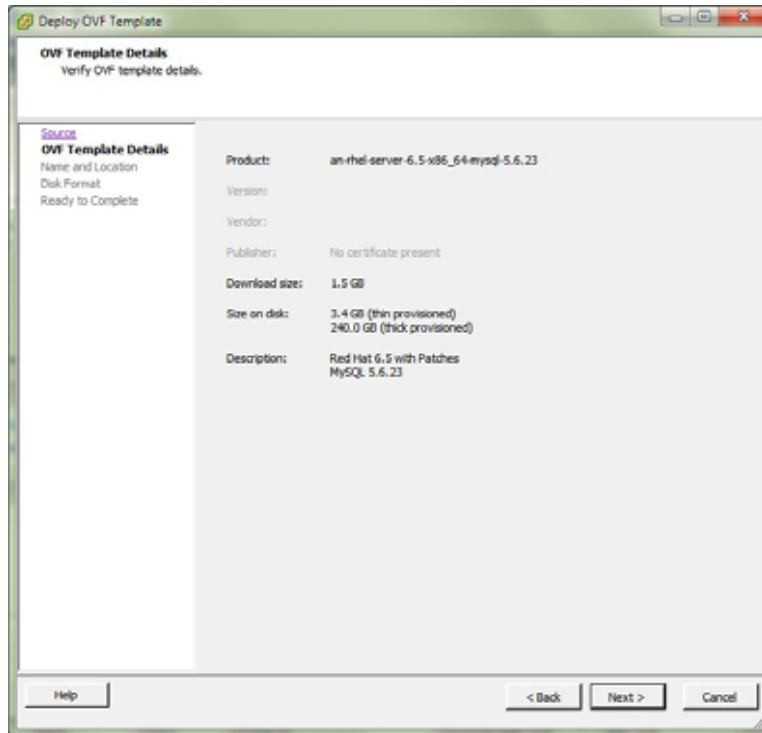
- 1 Download the OVA (an-pls-v1.3-rhel-server-6.8-x86\_64.ova) from Affirmed Networks and save it to the Windows system where the vSphere or vCenter client is located.
- 2 To start the vSphere client, select **Start > Programs > VMware > VMware vSphere Client** and log in.
- 3 From the **Configuration** tab, select **File > Deploy OVF Template**.

**Figure 2-43 Source Screen**



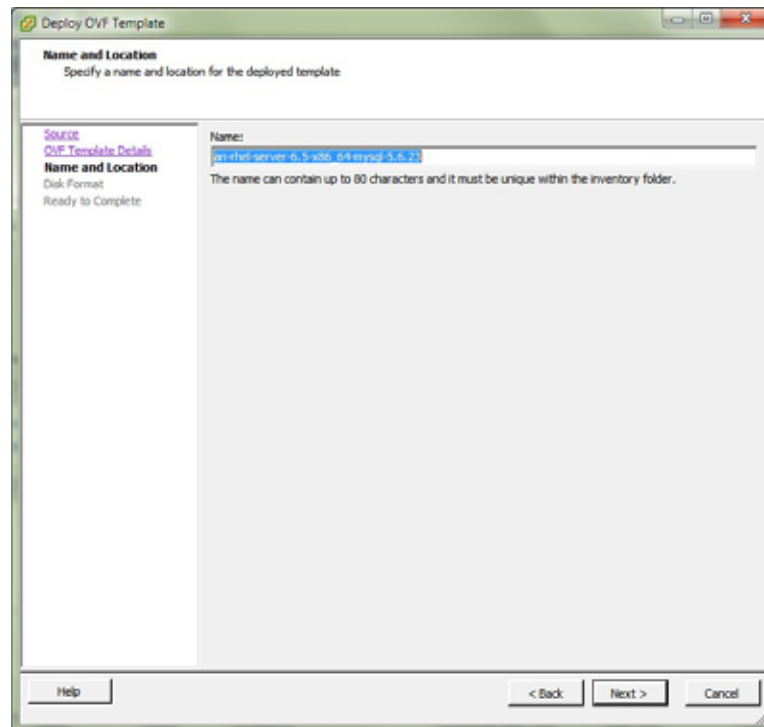
- 4 On the Source screen:
  - a Click **Browse** and select the OVA downloaded from Affirmed Networks.
  - b Click **Open**.
  - c Click **Next**.

Figure 2-44 Verification Screen



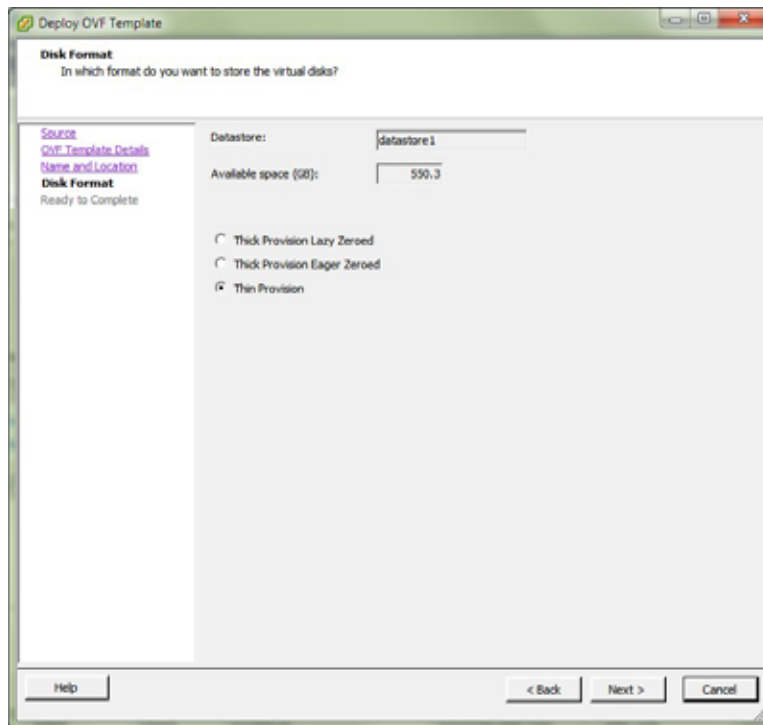
- 5 On the Verification screen, click **Next**.



**Figure 2-45 Name and Location Screen**

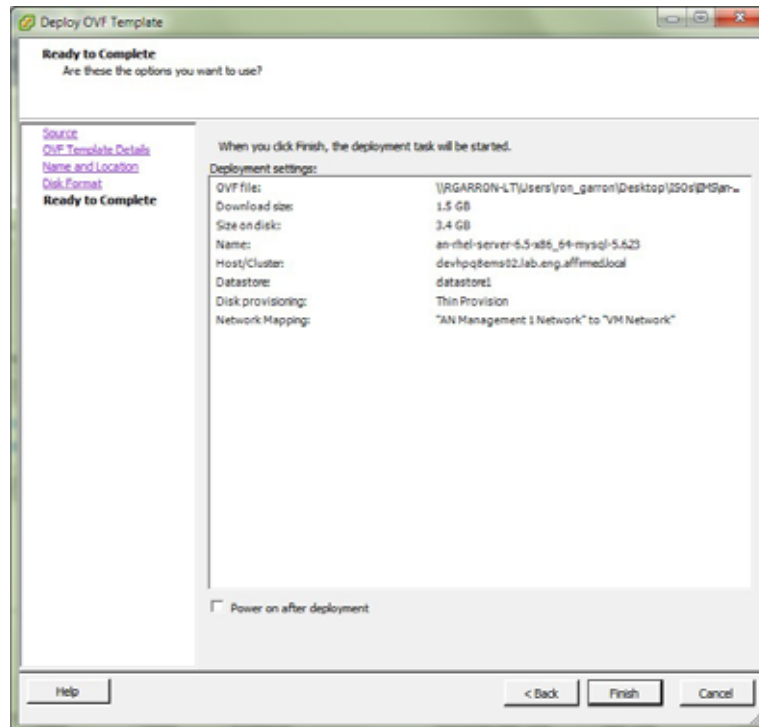
- 6 On the Name and Location screen, enter a name for the VM. Click **Next**.

Figure 2-46 Disk Format Screen



- 7 On the Disk Format screen, select **Thin Provision**. Click **Next**.

Figure 2-47 Ready to Complete Screen



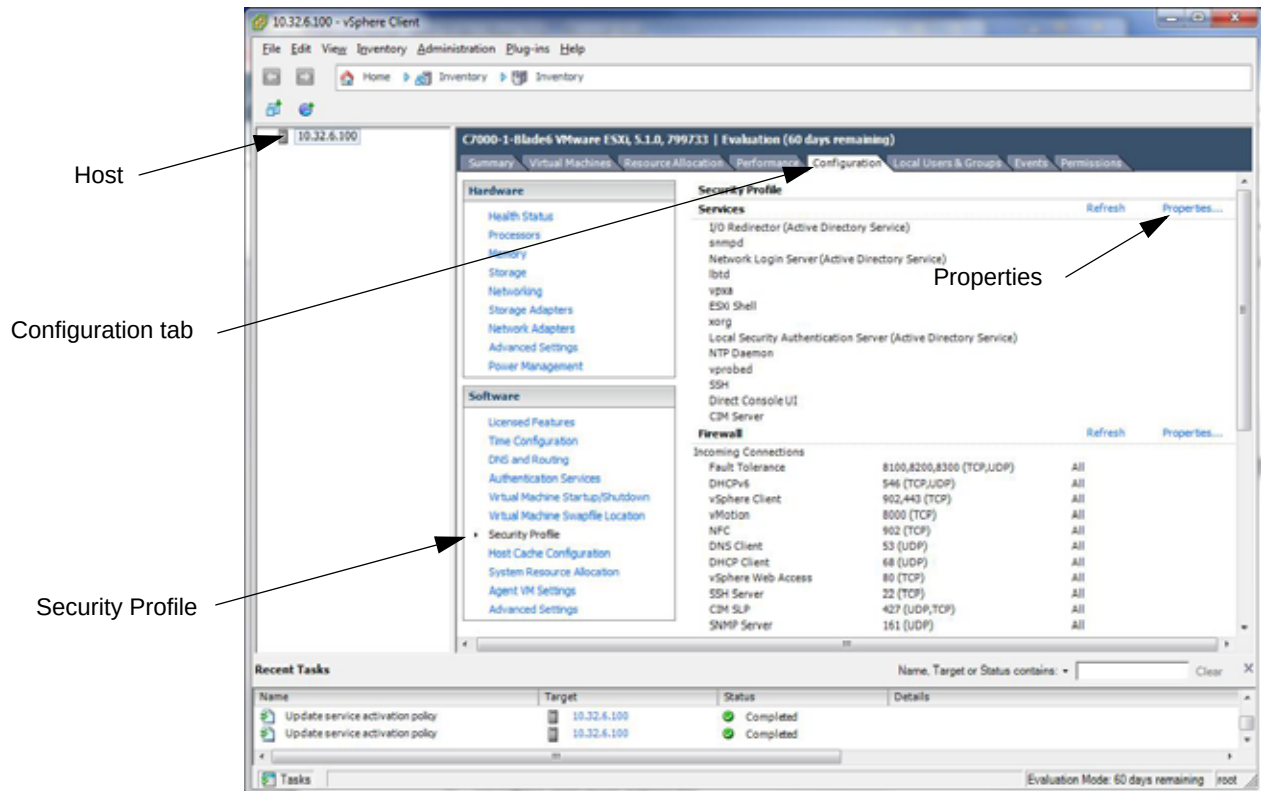
- 8 On the Ready to Complete screen, unmark the **Power on after deployment** checkbox. Click **Finish**.
- 9 On the Completed Successfully screen, click **Close**.

## Configure Security Profile

To configure the security profile:

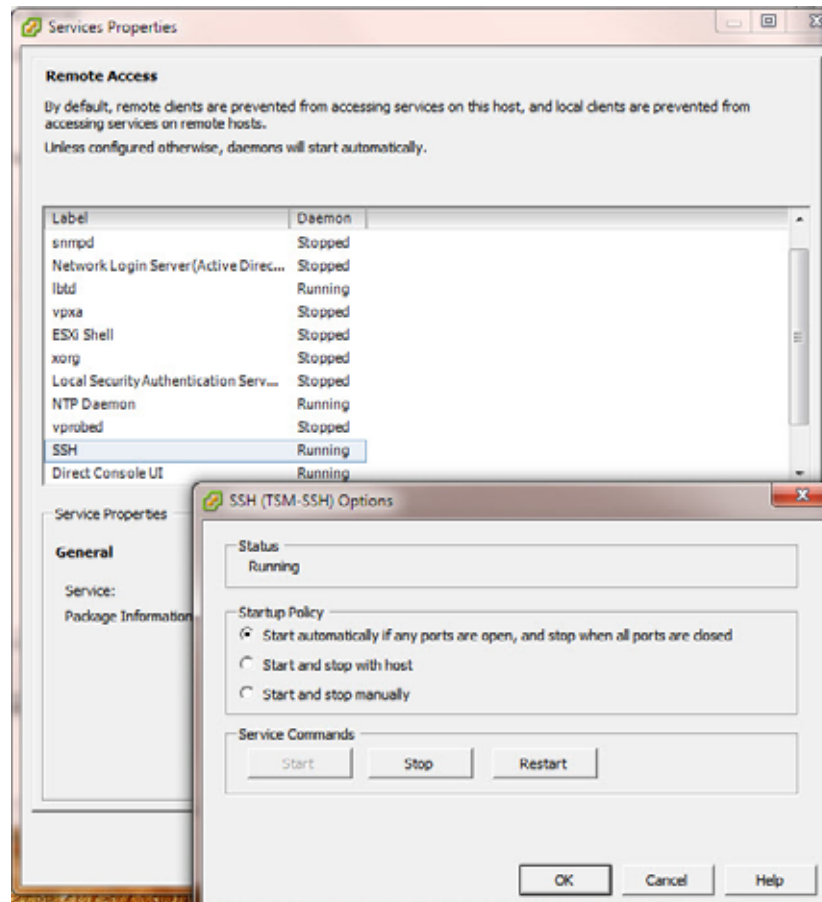
- 1 In vSphere:
  - a Click the host.
  - b Click the **Configuration** tab.
  - c In the **Software** list, click **Security Profile**.
  - d Under Services, click **Properties**.

Figure 2-48 Security Profile



- 2 To enable SSH:
  - a Click **SSH**.
  - b Click **Options**.
  - c Mark **Start Automatically**.
  - d Click **OK**.

Figure 2-49 SSH Settings

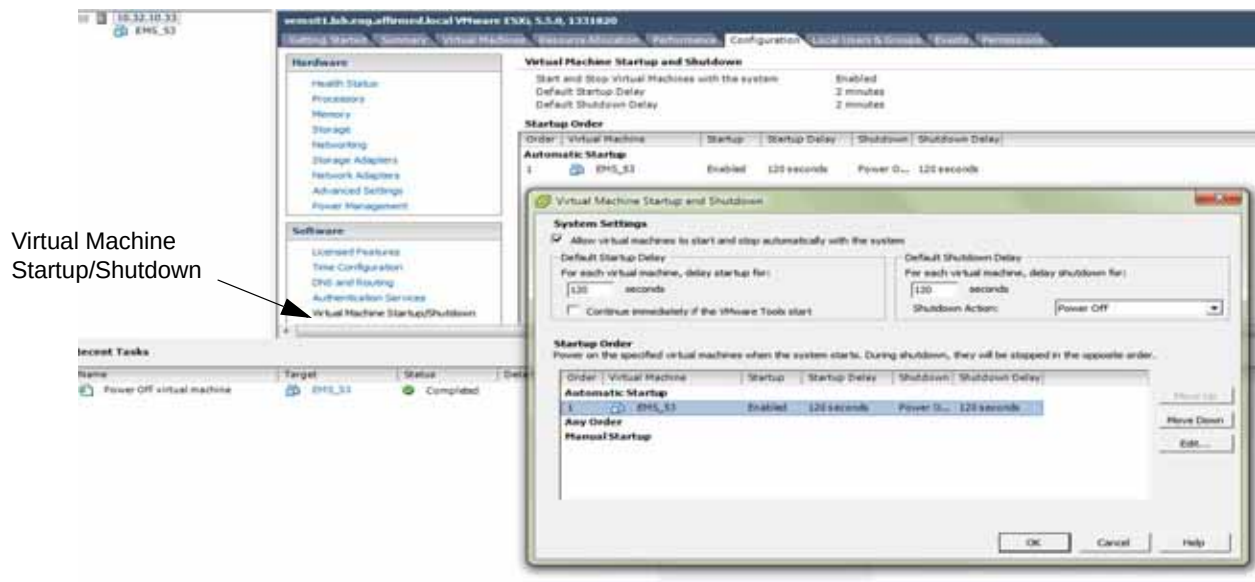


## Configure the PLS VM to Automatically Start Up

To configure the PLS VM to automatically start up:

- 1 If the PLS guest VM is not setup to automatic start up when the host is started, perform these steps:
  - a Click the host.
  - b Click the **Configuration** tab.
  - c In the **Software** list, click **Virtual Machine Startup/Shutdown**.
  - d Click **Properties**.

Figure 2-50 Virtual Machine Startup/Shutdown



- e Mark the **Allow virtual machines to start and stop automatically with the machine** checkbox.
- f In **Startup Order**, click the PLS guest VM and click **Move Up** to move it to Automatic Startup.
- g Click **OK**.

## Complete the Interface and DNS Setup

To complete the interface and DNS setup:

- 1 Power up the PLS VM.
- 2 From the console, log in as root (default password affirmedPLS).

The system prompts whether you would like to set up the server:

PLS is uninitialized, would you like to set up the server <y/n>

- 3 Enter y.

The system prompts to enter PLS commissioning information:

- System time
- PLS host name
- PLS management IP address
- PLS management subnet mask (IPv4 only)

- PLS management prefix (*IPv6 only*)
  - PLS management gateway IP address
  - PLS management DNS server
  - PLS management DNS search path
- 4 Enter each value.
  - 5 At the confirmation prompt, enter **y**.

The system is now ready for PLS installation. Proceed to [Standalone Systems](#).

## KVM System

This section describes how to prepare a KVM virtual machine, including:

- [Install the qcow2 Disk Image](#)
- [Complete the Interface and DNS Setup](#)

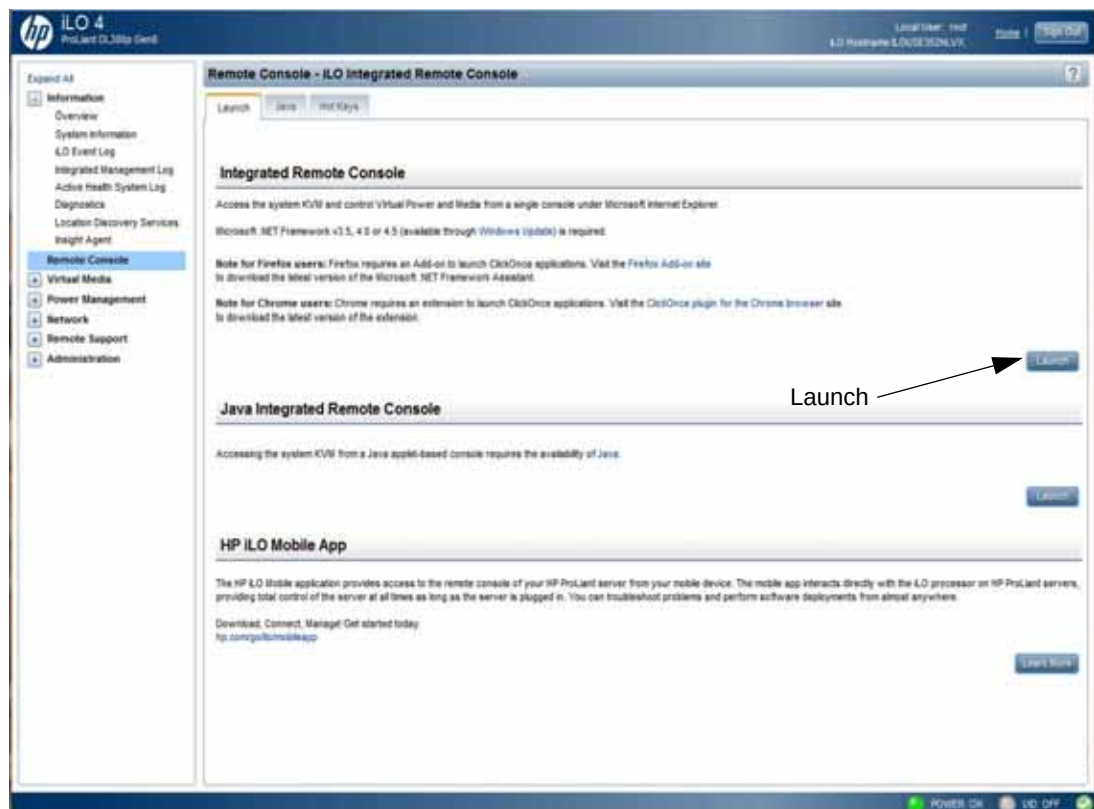
This procedure requires you to have KVM installed and configured. For instructions, see the *MCC Deployment Guide* for your virtual platform.

### Install the qcow2 Disk Image

To install the qcow2 disk image:

- 1 Download the qcow2 disk image from Affirmed Networks:  
an-pls-v1.3-rhel-server-6.8-x86\_64.qcow2
- 2 From a browser, log onto the iLO and click **Launch** to access the Remote Console (Firefox browser requires a Firefox Add-on to launch the Remote Console).

Figure 2-51 iLO

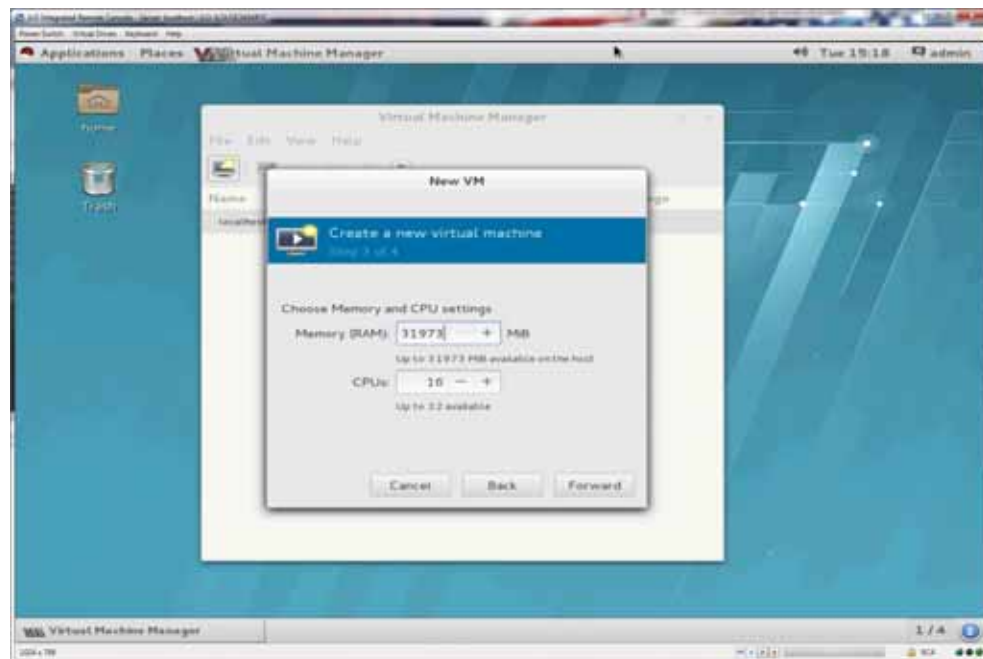


- 3 As root (default password affirmedPLS), select **Places > Computer** from the iLO Remote Console menu.
- 4 Navigate to /home/images.



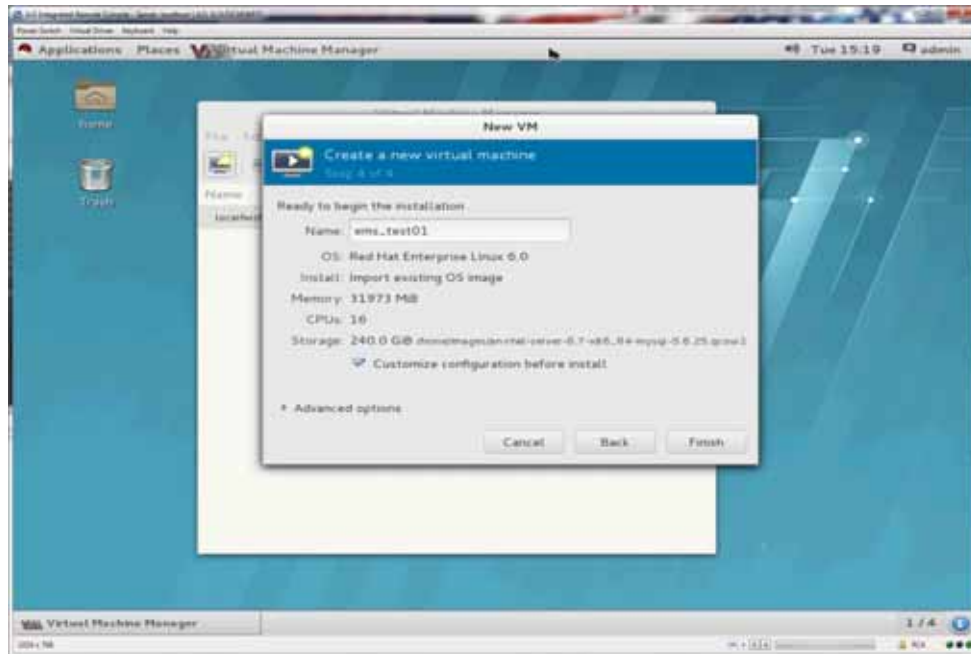
- 5 Right-click the directory and select **Open in Terminal**.
- 6 Copy the qcow2 disk image file into home/images.
- 7 From the iLO Remote Console menu, select **Applications > System Tools > Virtual Machine Manager**.
- 8 From the Virtual Machine Manager menu, select **File > New Virtual Machine**.
- 9 Click **Import Existing Disk Image**.
- 10 Click **Browse** and navigate to /home/images for the location of the qcow2 image.
- 11 In the OS Type field, select **Linux**.
- 12 In the Version field, select **Red Hat Enterprise Linux 6.0**.
- 13 Click **Forward**.
- 14 In the Memory (RAM) field, select **32768**.
- 15 In the CPUs field, select **16**.
- 16 Click **Forward**.

Figure 2-52 Memory and CPUs



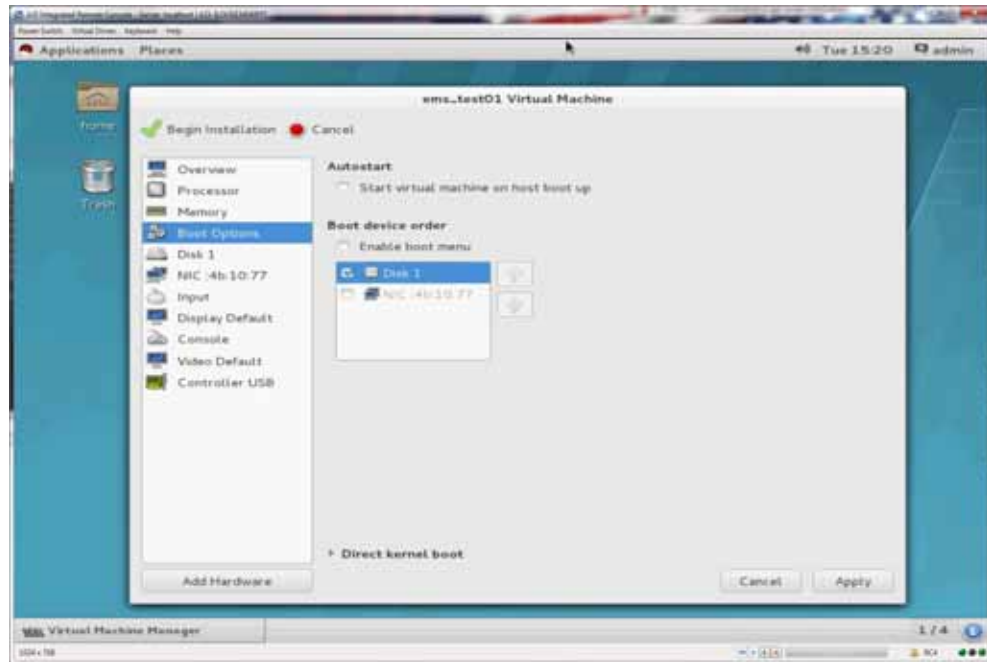
- 17 In the Name field, enter a name for the VM.
- 18 Mark the **Customize configuration before install** checkbox.
- 19 Click **Finish**.

**Figure 2-53 Customize Configuration Before Install**



- 20 Highlight Boot Options.
- 21 In the Boot Device Order field, select Disk 1.
- 22 Click **Apply**.

**Figure 2-54 Boot Options**



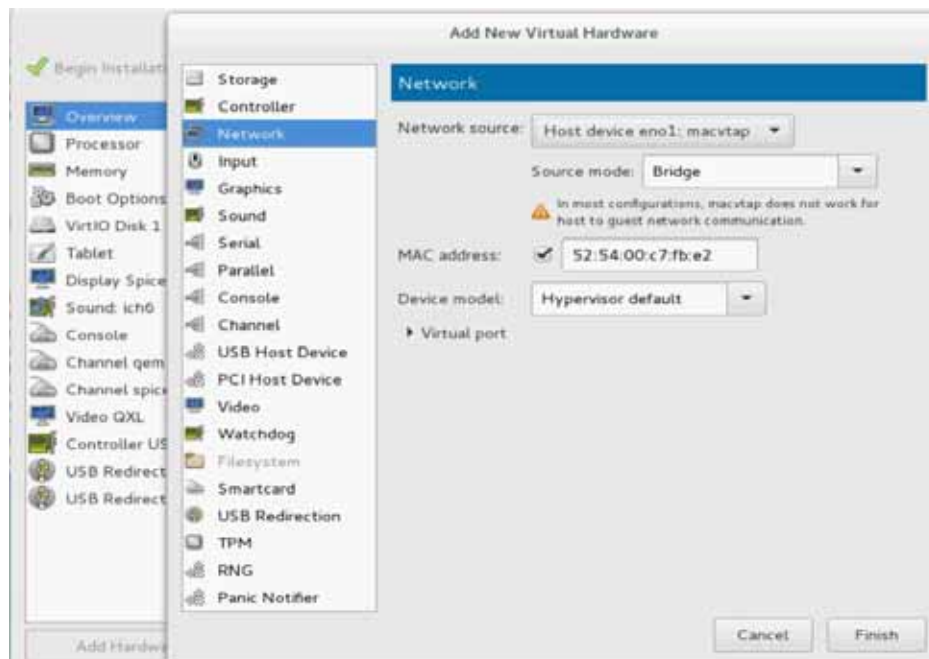
- 23 Click **Add Hardware** to add a network card.

**Figure 2-55 Add a Network Card**



- 24 Click **Network**.
- 25 In the Network Source field, select Host device eno1: macvtap.
- 26 Click **Finish**.

**Figure 2-56 Add New Virtual Hardware**



## Complete the Interface and DNS Setup

### To complete the interface and DNS setup:

- 1 Power up the PLS VM.
- 2 From the console, log in as root (default password affirmedPLS).  
The system prompts whether you would like to set up the server:  
`PLS is uninitialized, would you like to set up the server <y/n>`
- 3 Enter **y**.  
The system prompts to enter PLS commissioning information:
  - System time
  - PLS host name
  - PLS management IP address
  - PLS management subnet mask (*IPv4 only*)
  - PLS management prefix (*IPv6 only*)
  - PLS management gateway IP address
  - PLS management DNS server
  - PLS management DNS search path
- 4 Enter each value.
- 5 At the confirmation prompt, enter **y**.  
The system is now ready for PLS installation. Proceed to [Standalone Systems](#).



## Standalone Systems

This chapter describes how to install the PLS server and database software.

### Fresh Installation

#### Preparing for the Installation

**If you need to reconfigure the IP address of the PLS server:**

- 1 Change the PLS server IP address in the following file:  
*/etc/sysconfig/network-scripts/ifcfg-eth0*
- 2 Reboot the server or enter the following command:  
**service network restart**
- 3 Add the PLS server hostname and IP address to the following file:  
*/etc/sysconfig/network*
- 4 Reboot the server.

## Installation Instructions

---

**Note:** *If the PLS server is a factory-built system, skip this section.*

---

You must download the PLS software package from the SFTP server. See Affirmed Networks for instructions on how to access the SFTP server.

### Bare Metal System

**To fresh install the PLS software on a Bare Metal System:**

- 1 Execute the following command:

```
rpm -ivh AffirmedPlsBm-release-5.x86_64.rpm
```

(Note: include path of the <AffirmedPlsBm-release-5.x86\_64.rpm>)

where *release* is in the form of x.x.x.x. For example:

AffirmedPlsBm-7.0.0.0-5.x86\_64.rpm

Once you have installed the PLS software, you can start the server. For instructions, see [Standalone PLS Server Startup](#).

### Virtual Machine

**To fresh install the PLS software on a Virtual Machine:**

- 1 Execute the following command:

```
rpm -ivh AffirmedPls-release-5.x86_64.rpm
```

(Note: include path of the <AffirmedPls-release-5.x86\_64.rpm>)

where *release* is in the form of x.x.x.x. For example:

AffirmedPls-7.0.0.0-5.x86\_64.rpm

Once you have installed the PLS software, you can start the server. For instructions, see [Standalone PLS Server Startup](#).



## Other Administrative Information

This chapter provides the following administrative information:

- [PLS Server Administration Scripts](#) — Describes the various scripts you use to administer the PLS server:
  - [Standalone PLS Server Startup](#)
  - [Standalone PLS Server Shutdown](#)
  - [PLS Server Status](#)
- [NTP Server Configuration](#)
- [Running PLS as a Non-root User](#)
- [Changing Hostname/IP Address of the PLS Server](#)
- [Changing Timezone of the PLS Server](#)
- [Increasing PLS Server Resources on a Virtual Machine](#)
- [Optional: Channel Bonding on PLS Physical Interfaces](#)
- [Ports Used by the PLS System](#)

## PLS Server Administration Scripts

### Standalone PLS Server Startup

Use the following command to start the PLS server:

```
pls start
```

### Standalone PLS Server Shutdown

Use the following command to shut down the PLS server:

```
pls stop
```

### PLS Server Status

Use the following command to determine the current status of the PLS server:

```
pls status
```

PLS displays the server status:

```
[root@NewPLS ~]# pls status
17519 pts/0    Sl      0:00 /opt/Affirmed/PLS/bin/LmPlsOamServer
17531 ?          S       0:02 ../bin/lserv -s affirmed1
[root@NewPLS ~]#
```

## NTP Server Configuration

A Network Timing Protocol server is required to synchronize timing between PLS and the AN devices.

### Bare Metal Systems

Use the `configure_ntp.sh` script to configure a system NTP server and enable its administrative state.

**To configure a system NTP server and enable its administrative state:**

- 1 Enter the following commands:

```
cd /opt/Affirmed/PLS/bin
./configure_ntp.sh
```

- 2 When prompted, enter the IP address of the NTP time server.

The script updates the NTP configuration file and brings up the configured NTP server.

- 3 Enter the following command to verify that the NTP server is successfully started:

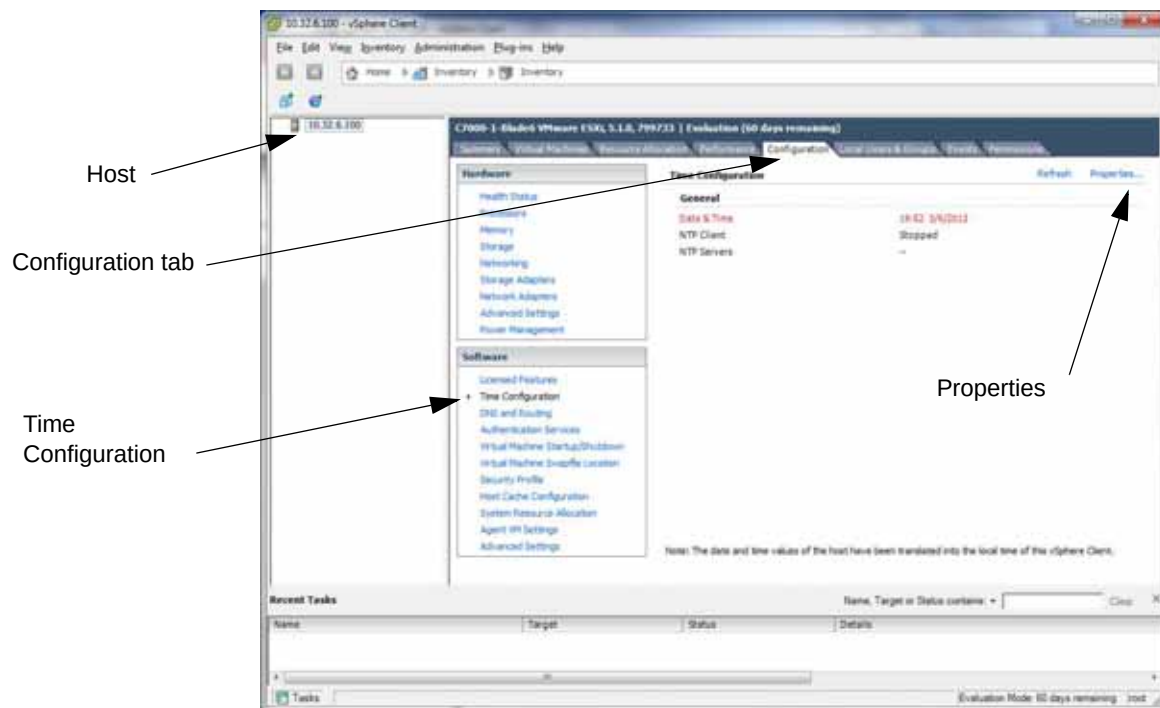
```
ps -ef | grep -i ntpd
ntp      13760      1  0 10:59 ?          00:00:00 ntpd -u ntp:ntp -p
/var/run/ntpd.pid -g
```

## Virtual Machine

### To enable NTP:

- 1 To start the vSphere client, select **Start > Programs > VMware > VMware vSphere Client**.

Figure 4-1 Time Configuration



- 2 Perform the following steps at the initial screen:
  - a Click the host.
  - b Click the **Configuration** tab.
  - c In the **Software** list, click **Time Configuration**.
  - d Click **Properties**.
- 3 In Time Configuration:
  - a Set the Time and Date for the host.
  - b Mark the **NTP Client Enabled** checkbox and click **Options**.
- 4 In the NTP daemon:
  - a Click **NTP Settings**.
  - b Click **Add**.
  - c Enter an NTP server IP address.
  - d Mark the **Restart NTP service to apply changes** checkbox.

- e Click **OK**.
  - 5 In the NTP daemon:
    - a Click **General**.
    - b Click **Start**.
    - c Click **OK**.
    - d In Time Configuration, click **Options**.
    - e Mark the **Start and stop with host** checkbox and click **Start**.
    - f Click **OK**.
  - 6 Verify that all settings are listed correctly in the Time Configuration screen.

## Running PLS as a Non-root User

PLS can be run as either the root user or a non-root user. However, the non-root user requires additional configuration to bind to required privileged ports such as ports 80 and 443 for HTTPS sessions and port 162 for receiving SNMP traps.

This section describes how to provide the appropriate privileges to non-root users so that they can use the scripts to start, stop, show status, and perform other PLS maintenance operations.

## Add Entries to the `/etc/sudoers` File

You need to add entries to the `/etc/sudoers` file to allow listed users to run PLS scripts as sudo.

**To add entries to the sudoers file:**

- 1 Issue the following command to edit the sudoers file:

**# visudo**

The content of the file is as follows:

```
## Sudoers allows particular users to run various commands as
## the root user, without needing the root password.
##
## Examples are provided at the bottom of the file for collections
## of related commands, which can then be delegated out to particular
## users or groups.
##
## This file must be edited with the 'visudo' command.

## Host Aliases
## Groups of machines. You may prefer to use hostnames (perhaps using
## wildcards for entire domains) or IP addresses instead.
# Host_Alias    FILESERVERS = fs1, fs2
# Host_Alias    MAILSERVERS = smtp, smtp2

## User Aliases
## These aren't often necessary, as you can use regular groups
## (ie, from files, LDAP, NIS, etc) in this file - just use %groupname
## rather than USERALIAS
# User_Alias ADMINS = jsmith, mikem
```

```
## Command Aliases
## These are groups of related commands...

## Networking
# Cmnd_Alias NETWORKING = /sbin/route, /sbin/ifconfig, /bin/ping,
/sbin/dhclient, /usr/bin/net, /sbin/iptables, /usr/bin/rfcomm,
/usr/bin/wvdial, /sbin/iwconfig, /sbin/mii-tool

## Installation and management of software
# Cmnd_Alias SOFTWARE = /bin/rpm, /usr/bin/up2date, /usr/bin/yum

## Services
# Cmnd_Alias SERVICES = /sbin/service, /sbin/chkconfig

## Updating the locate database
# Cmnd_Alias LOCATE = /usr/bin/updatedb

## Storage
# Cmnd_Alias STORAGE = /sbin/fdisk, /sbin/sfdisk, /sbin/parted,
/sbin/partprobe, /bin/mount, /bin/umount

## Delegating permissions
# Cmnd_Alias DELEGATING = /usr/sbin/visudo, /bin/chown, /bin/chmod,
/bin/chgrp

## Processes
# Cmnd_Alias PROCESSES = /bin/nice, /bin/kill, /usr/bin/kill,
/usr/bin/killall

## Drivers
# Cmnd_Alias DRIVERS = /sbin/modprobe

# Defaults specification

#
# Disable "ssh hostname sudo <cmd>", because it will show the password
in clear.
#         You have to run "ssh -t hostname sudo <cmd>".
#
Defaults    requiretty
```

```
#
# Refuse to run if unable to disable echo on the tty. This setting
# should also be
# changed in order to be able to use sudo without a tty. See requiretty
# above.
#
Defaults    !visiblepw

#
# Preserving HOME has security implications since many programs
# use it when searching for configuration files. Note that HOME
# is already set when the the env_reset option is enabled, so
# this option is only effective for configurations where either
# env_reset is disabled or HOME is present in the env_keep list.
#
Defaults    always_set_home

Defaults    env_reset
Defaults    env_keep = "COLORS DISPLAY HOSTNAME HISTSIZE INPUTRC
                        KDEDIR LS_COLORS"
Defaults    env_keep += "MAIL PS1 PS2 QTDIR USERNAME LANG LC_ADDRESS
                        LC_CTYPE"
Defaults    env_keep += "LC_COLLATE LC_IDENTIFICATION LC_MEASUREMENT
                        LC_MESSAGES"
Defaults    env_keep += "LC_MONETARY LC_NAME LC_NUMERIC LC_PAPER
                        LC_TELEPHONE"
Defaults    env_keep += "LC_TIME LC_ALL LANGUAGE LINGUAS _XKB_CHARSET
                        XAUTHORITY"

#
# Adding HOME to env_keep may enable a user to run unrestricted
# commands via sudo.
#
# Defaults    env_keep += "HOME"

Defaults    secure_path = /sbin:/bin:/usr/sbin:/usr/bin

## Next comes the main part: which users can run what software on
## which machines (the sudoers file can be shared between multiple
## systems).
## Syntax:
##
```



```

##      user      MACHINE=COMMANDS
##
## The COMMANDS section may have other options added to it.
##
## Allow root to run any commands anywhere
root    ALL=(ALL)        ALL

## Allows members of the 'sys' group to run networking, software,
## service management apps and more.
# %sys ALL = NETWORKING, SOFTWARE, SERVICES, STORAGE, DELEGATING,
PROCESSES, LOCATE, DRIVERS

## Allows people in group wheel to run all commands
# %wheel      ALL=(ALL)        ALL

## Same thing without a password
# %wheel      ALL=(ALL)        NOPASSWD: ALL

## Allows members of the users group to mount and unmount the
## cdrom as root
# %users ALL=/sbin/mount /mnt/cdrom, /sbin/umount /mnt/cdrom

## Allows members of the users group to shutdown this system
# %users localhost=/sbin/shutdown -h now

## Read drop-in files from /etc/sudoers.d (the # here does not mean a
comment)
#includedir /etc/sudoers.d

```

- 2 You can provide access to the PLS maintenance scripts from the pre-configured **affirmed** user (belonging to the **affirmed** group) or you can provide access to the **affirmed** group.

**Note:** You can replace the user and user group in this step with other users and groups.

- To provide access from the **affirmed** user account, append the following line to the end of the file. Enter it as a single line:

```

affirmed ALL=(ALL) NOPASSWD: /opt/Affirmed/NMS/bin/emsstatus,
/opt/Affirmed/NMS/bin/starttems.sh, /opt/Affirmed/NMS/bin/emsstop,
/opt/Affirmed/NMS/bin/emsha, /opt/Affirmed/NMS/bin/emsupgrade,
/opt/Affirmed/NMS/bin/configure_dr,
/opt/Affirmed/NMS/bin/configure_ha,
/opt/Affirmed/NMS/bin/emslogdump.pl,
/opt/Affirmed/NMS/bin/configure_ntp.sh,
/opt/Affirmed/NMS/bin/haupgrade,
/opt/Affirmed/NMS/bin/configureserver.sh

```

- To provide access from the **affirmed** group, append the following line to the file with a % before the group name. Enter it as a single line:

```
%affirmed ALL=(ALL) NOPASSWD: /opt/Affirmed/NMS/bin/emsstatus,  
/opt/Affirmed/NMS/bin/starttems.sh, /opt/Affirmed/NMS/bin/emsstop,  
/opt/Affirmed/NMS/bin/emsha, /opt/Affirmed/NMS/bin/emsupgrade,  
/opt/Affirmed/NMS/bin/configure_dr,  
/opt/Affirmed/NMS/bin/configure_ha,  
/opt/Affirmed/NMS/bin/emslogdump.pl,  
/opt/Affirmed/NMS/bin/configure_ntp.sh,  
/opt/Affirmed/NMS/bin/haupgrade,  
/opt/Affirmed/NMS/bin/configureserver.sh
```

- 3 Save the sudoers file.

The PLS maintenance scripts can now be executed by non-root users. Proceed to [Run PLS Scripts as a Non-root User](#).

## Run PLS Scripts as a Non-root User

The following PLS maintenance scripts can be executed by non-root users using the `sudo` command option:

- `emsstatus`
- `starttems.sh`
- `emsstop`
- `emsha`
- `emsupgrade`
- `configure_dr`
- `configure_ha`
- `emslogdump.pl`
- `configure_ntp.sh`
- `haupgrade`
- `configureserver.sh`

### To run PLS scripts as a non-root user:

- 1 Log in as the `affirmed` user (or other user to which you added privileges):
- 2 Issue the following command:  
`cd /opt/Affirmed/PLS/bin`
- 3 Run the script with the `sudo` option. For example:  
`sudo ./emsstatus`

```
***** PLS Server Status *****
```

```
Node: qahpems02 - 10.6.1.69
```

```
EMS Package: AffirmedEms-4.2.2.20-341
```

```
EMS Process: Running [3261]
```

```
MySQL
```

```
Database: Running
```

```
Server Id: 1
```

```
*****
```

## Changing Hostname/IP Address of the PLS Server

This section describes how to change the hostname and/or IP address on the PLS server.

### To change the IP address of the PLS server:

- 1 Stop PLS:  
`pls stop`
- 2 Change the PLS server IP address in the following file:  
`/etc/sysconfig/network-scripts/ifcfg-eth0`
- 3 Reboot the server or enter the following command:  
`service network restart`
- 4 Add the PLS server hostname and IP address to the following file:  
`/etc/sysconfig/network`
- 5 Restart the server:  
`pls start`

### To change the Hostname of the PLS server:

- 1 Stop PLS:  
`pls stop`
- 2 Remove the old entries from `/etc/hosts`.
- 3 Change hostname using the following command:  
`hostname <new_hostname>`
- 4 Change hostname in `/etc/sysconfig/network` file.
- 5 Restart the server:  
`service network restart`
- 6 Restart PLS:  
`pls start`

## Changing Timezone of the PLS Server

Changing time on PLS server will impact licenses, because the license system detects the change and considers it as time tampering.

### Prerequisite

If the PLS server is running on a virtual machine, you must have NTP enabled on the host. For instructions, see [Virtual Machine](#).

### Procedure for a Standalone System

**To change the timezone of the PLS Server:**

- 1 Move the existing local time file in case you need it later:  

```
# cd /etc/
# ls -l | grep localtime
-rw-r--r--. 1 root root 3519 Jul 14 14:06 localtime
# mv localtime localtime.old
```
- 2 Issue the following commands to create a link for a new local time file and link it to the timezone you want from the available zones under /usr/share/zoneinfo/:

---

**Note:** This example sets the time zone to pacific time. Change your command accordingly.

---

```
# ln -s /usr/share/zoneinfo/PST8PDT localtime
# ls -l localtime
lrwxrwxrwx 1 root root 27 Sep 17 11:21 localtime ->
/usr/share/zoneinfo/PST8PDT
```

- 3 Issue the following commands to edit the clock file from /etc/sysconfig/clock and set it to the matching zone as the local time file:  

```
# vi /etc/sysconfig/clock
# cat /etc/sysconfig/clock
ZONE="America/Los_Angeles"
```
- 4 Reboot the PLS Server:  

```
# reboot
```

When the server comes back up, it is using the new timezone.

## Procedure for a Cluster

This section describes how to change timezone in an HA + DR cluster.

### To change the timezone of the cluster:

- 1 On the master pair, issue the following commands to shutdown the standby server and the active server, in that order:

#### Standby server:

```
cd /opt/Affirmed/NMS/bin
./emsha stopAll
```

#### Active server:

```
cd /opt/Affirmed/NMS/bin
./emsha stopAll
```

- 2 At each confirmation prompt, enter **y**.
- 3 No action is required on the slave pair, as long as no PLS process is running. Issue the following command on all servers in the cluster to ensure that no PLS process is running.

```
cd /opt/Affirmed/NMS/bin
./emsstatus
```

- 4 Move the existing local time file in case you need it later:

```
# cd /etc/
# ls -l | grep localtime
-rw-r--r--. 1 root root 3519 Jul 14 14:06 localtime
# mv localtime localtime.old
```

- 5 Issue the following commands to create a link for a new local time file and link it to the timezone you want from the available zones under /usr/share/zoneinfo/:

---

**Note:** This example sets the time zone to pacific time. Change your command accordingly.

---

```
# ln -s /usr/share/zoneinfo/PST8PDT localtime
# ls -l localtime
lrwxrwxrwx 1 root root 27 Sep 17 11:21 localtime ->
/usr/share/zoneinfo/PST8PDT
```

- 6 Issue the following commands to edit the clock file from /etc/sysconfig/clock and set it to the matching zone as the local time file:

```
# vi /etc/sysconfig/clock
# cat /etc/sysconfig/clock
ZONE="America/Los_Angeles"
```

- 7 On the master pair, issue the following commands to start the active server and the standby server, in that order:

**Active server:**

```
/opt/Affirmed/NMS/bin % ./emsha startAll
```

**Standby server:**

```
/opt/Affirmed/NMS/bin % ./emsha startAll
```

- 8 At each confirmation prompt, enter **y**.
- 9 When each system returns the success message:

```
Server started successfully
```

Press **Enter** to return to the operating system prompt.

When the servers come back up, they are using the new timezone.

- 10 Issue the following command on all servers in the cluster to ensure that everything is operational.

```
cd /opt/Affirmed/NMS/bin  
./emsstatus
```

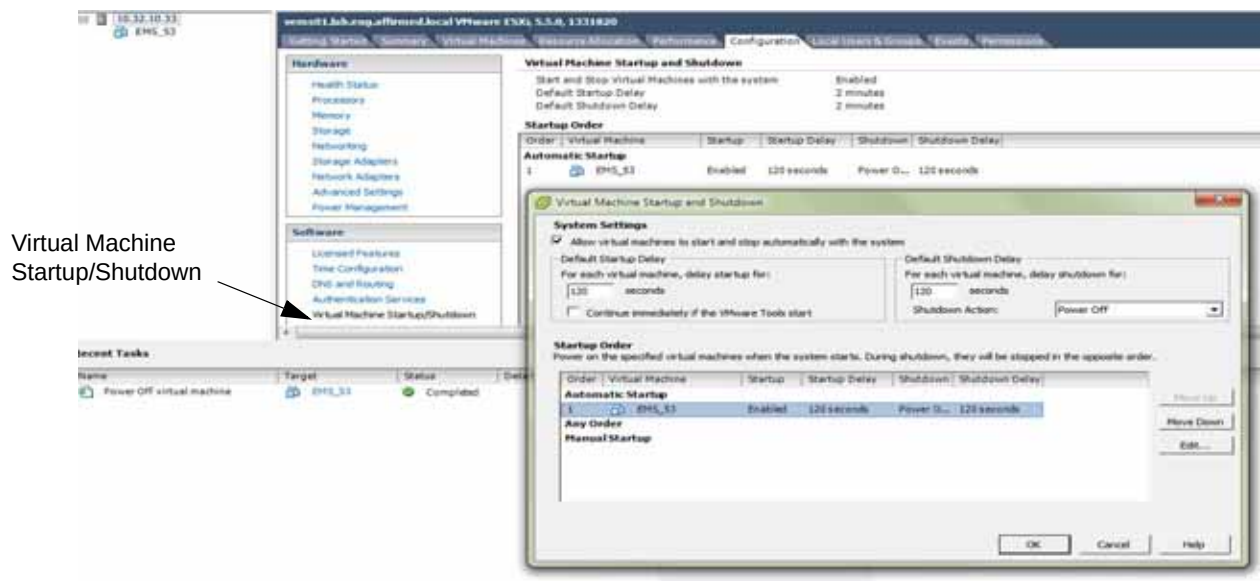
## Increasing PLS Server Resources on a Virtual Machine

This section describes how to increase the number of CPUs used by the PLS VM.

**To increase the number of CPUs used by the PLS VM:**

- 1 Stop the PLS server:  
**pls stop**
- 2 Select **Start > Programs > VMware > VMware vSphere Client**.
- 3 Under Host, right-click the PLS guest VM and select **Power > Power Off**.
- 4 If the PLS guest VM is not setup to automatic start up when the host is started, perform these steps:
  - a Click the **Configuration** tab.
  - b In the **Software** list, click **Virtual Machine Startup/Shutdown**.
  - c Click **Properties**.

Figure 4-2 Virtual Machine Startup/Shutdown

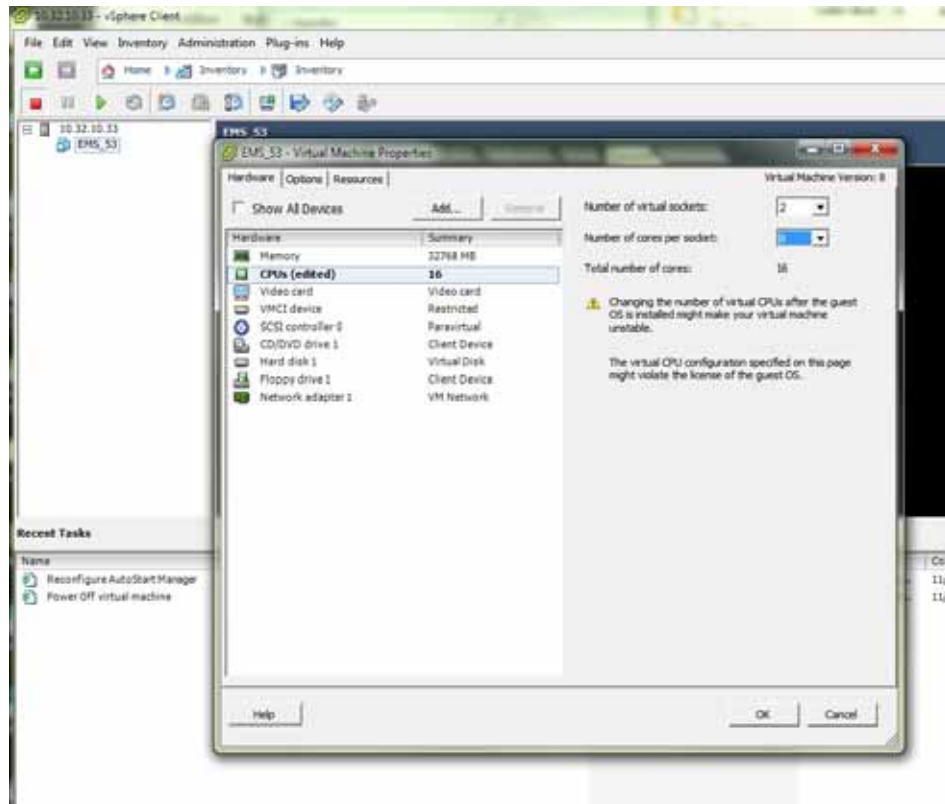


- d Mark the **Allow virtual machines to start and stop automatically with the machine** checkbox.
  - e In **Startup Order**, click the PLS guest VM and click **Move Up** to move it to Automatic Startup.
  - f Click **OK**.
- 5 To change the CPU setting for the PLS guest VM, perform these steps:
  - a Right-click PLS guest VM and select **Edit Settings**.



- b Click the **Hardware** tab.
- c Click **CPUs**.
- d Keep the number of virtual sockets at 2. Increase the number of virtual cores to 8, so the total number of cores is 16.
- e Click **OK**.

**Figure 4-3 CPUs Settings**



- 6 Under Host, right-click the PLS guest VM and select **Power > Power On**.
- 7 Log onto the PLS server as **root** (default password affirmedPLS).
- 8 Start the PLS server:

**Standalone Server:**

**pls stop**

**2-node HA Cluster:**

Issue the following commands from one of the servers in the cluster:

```
cd /opt/Affirmed/NMS/bin
./emsha startAll primary
./emsha startAll secondary
```

**4-node HA Cluster:**

Issue the following commands from one of the servers in the master pair:

```
cd /opt/Affirmed/NMS/bin  
./emsha startAll primary  
./emsha startAll secondary
```

- 9 At each confirmation prompt, enter **y**.
- 10 When each system returns the success message:  
Server started successfully  
Press **Enter** to return to the operating system prompt.

## Optional: Channel Bonding on PLS Physical Interfaces

Channel bonding enables multiple network interfaces to act as a single interface, thereby increasing bandwidth and providing redundancy. You can configure channel bonding to operate with PLS servers to remove a single point of failure for the PLS management network.

### To set up Channel Bonding:

- 1 To create a channel bonding interface, create a file in the `/etc/sysconfig/network-scripts/` directory named `ifcfg-bondN`, where *N* is the interface number.

- 2 Edit the `ifcfg-bondN` file and add the following lines:

```
DEVICE=bondN
IPADDR=<IP address of the device>
NETMASK=<Netmask>
ONBOOT=yes
BOOTPROTO=none
USERCTL=no
```

Where *N* is the interface number.

- 3 Save the file.
- 4 After the channel bonding interface is created, configure the network interfaces to be bound together by adding the MASTER and SLAVE directives to their corresponding configuration files. To do so, add the following lines to each network interface configuration file you are bonding.

```
ONBOOT=yes
MASTER=bondN
SLAVE=yes
USERCTL=no
mode=active-backup
```

For example, to bond `eth0` and `eth1`, add these lines to `/etc/sysconfig/network-scripts/ifcfg-eth0` and `/etc/sysconfig/network-scripts/ifcfg-eth1`

- 5 As root (default password `affirmedPLS`), create a new file in the `/etc/modprobe.d/` directory named `<bonding>.conf`. It can have any filename with the required `.conf` extension. Add the following line to the file:

```
alias bondN bonding
```

Where *N* is the interface number, such as 0.

- 6 Execute the following commands to restart the network:

```
service network restart
```

## Ports Used by the PLS System

The PLS system uses the ports listed in the following table.

**Table 4-1** Ports Used by the PLS System

| TCP Port | UDP Port | Used by                                 |
|----------|----------|-----------------------------------------|
| 22       |          | PLS system Command Line Interface (CLI) |
|          | 5093     | PLS system                              |
| 4560     |          | PLS system                              |